

# Dancing on the Saddles: A Geometric Framework for Stochastic Equilibrium Dynamics<sup>†</sup>

Hanbaek Lee<sup>‡</sup>

*University of Cambridge*

August 2026

[\(click here for the latest version\)](#)

## Abstract

Extending saddle-path analysis to stochastic environments, I introduce *conditional saddle paths*: equilibrium trajectories under frozen exogenous regimes, traced by stochastic-equilibrium decision rules. I first use their *geometry* to diagnose state dependence: shock responses depend on position along the saddle and on slope gaps across regimes. I then show that their *monotonicity* delivers dimension reduction, since aggregate capital exactly indexes the equilibrium state within a frozen-regime spell in [Krusell and Smith \(1998\)](#). Across spells, a synchronization theorem bounds the residual gap between histories by the regimes they recently shared. Finally, extending the geometry to a two-dimensional conditional saddle *surface* with an investment externality, I analyze the optimal investment tax credit surface over the state space.

**Keywords:** Conditional saddle path, business cycles, state-dependent dynamics, sufficient statistics, heterogeneous agents, investment externality.

**JEL codes:** C62, D58, E32.

---

<sup>†</sup>I am grateful for helpful comments and discussions from Martin M. Andreasen, Yunus Aksoy, Marlon Azinovic-Yang, Adrien Bilal, Timo Boppart, Francisco Buera, Christopher Carroll, Vasco Carvalho, Minsu Chang, Rachel Leah Childers, Miguel H. Ferreira, Elisa Giannone, Matthias Hänsel, Tomohiro Hirano, Dirk Krueger, Andrea Lanteri, Benjamin Larin, Danial Lashkari, Tim Lee, Benjamin Moll, Juan Carlos Parra-Alvarez, Tommaso Porzio, Michael Reiter, Itay Saporta-Eksten, Yucheng Yang, Yuki Yao, and Jan Žemlička, as well as the seminar participants at Birkbeck, University of London, the University of Manchester, Aarhus University, QMUL E1 macro conference, and SED at Athens. All errors are my own.

<sup>‡</sup>Email: [hl610@cam.ac.uk](mailto:hl610@cam.ac.uk)

# 1 Introduction

Understanding equilibrium dynamics in models with aggregate uncertainty remains a central challenge in macroeconomics. Unlike deterministic models, where saddle-path diagrams provide immediate geometric intuition, stochastic equilibrium models lack comparable geometric frameworks. This makes it difficult to develop intuition about how economies respond to shocks, how different states interact, and why certain computational methods work. The challenge becomes particularly acute in heterogeneous-agent models, where the natural state variable is an infinite-dimensional wealth distribution, yet computational work routinely achieves dimension reduction to low-dimensional aggregates. Can we extend geometric saddle-path analysis to stochastic environments? What does such a framework reveal about equilibrium dynamics, state-dependent responses, and the success of computational approximations?

The paper makes three contributions, all drawn from a single geometric object. First, I define *conditional saddle paths*: equilibrium trajectories under frozen exogenous regimes, traced by the decision rules of the stochastic equilibrium. Each regime's trajectory draws a policy graph in the aggregate phase plane, decomposing fluctuations into movements *along* a conditional saddle (endogenous propagation) and *across* saddles (exogenous shocks), the stochastic analogue of the deterministic phase diagram. The geometry reveals all the possible dynamics within the equilibrium, both factual and counterfactual, without separate numerical results: the model's entire dynamics are visible in a single picture. A stark example is the shock response and its state dependence: impacts are state-independent when the saddles are parallel; relative tilt makes the same shock hit harder in some states; and even with parallel saddles, the response *shape* (peak timing and persistence) depends on position along the saddle. The graphs also visualize equilibrium-level comparative statics: a changed parameter (e.g., risk aversion) re-solves the recursive competitive equilibrium and tilts the saddles, showing how the parameter reshapes the equilibrium dynamics. Set against their perfect-foresight counterparts, the saddles yield a *flow anticipation wedge*: the area between the two families is the consumption (dis-)saved ahead of regime change, surplus geometry on the state space. Its welfare counterpart measures the value of anticipated reversion in consumption equivalents, visualizing the net surplus from the business cycle in the spirit of [Lucas \(1987\)](#). I develop these in a canonical RBC model (Sections 2–3).

Second, the same geometry speaks to a long-standing computational puzzle: why tracking a single aggregate can stand in for an entire wealth distribution. When an aggregate

variable moves monotonically along the conditional saddle, it exactly indexes the equilibrium state within a frozen-regime spell, so one number recovers the distribution and every price. I show that aggregate capital plays this role in [Krusell and Smith \(1998\)](#): within each spell the dimension reduction is exact along the entire conditional saddle. The key is the invariance of the phase diagram to heterogeneity: the heterogeneous economy's conditional  $K$ -nullclines coincide with those of the canonical RBC economy, because the zero-net-investment boundary is an accounting identity in aggregates, independent of *who* saves. This exact sufficiency is within-spell only: two histories can reach the same  $K$  carrying different distributions. Across spells, a synchronization theorem bounds this residual gap by the regimes the two histories recently shared. The contraction rate is anchored near the conditional steady state; the remaining discrepancy, small in [Krusell and Smith \(1998\)](#), is documented directly. Together, within-spell exactness and across-spell synchronization yield the near-global sufficiency of aggregate capital. The underlying monotonicity is verified two ways: a cheap order-preservation check on the computed frozen-regime map holds throughout this economy, and primitive bounds on MPCs and price feedback deliver it on an explicit region (Section 4.1).

Third, I extend the geometry to two predetermined states, where the conditional saddle becomes a *surface*. In an RBC economy where investment builds knowledge that firms do not pay for (Section 4.2), the planner traces a second surface, resting at higher capital and knowledge, and a state-contingent investment tax credit implements it. That credit is itself a surface over the two stocks, so optimal policy can be analyzed geometrically on the same state space. At the illustrative spillover it is 2.4 percent of gross investment, nearly proportional to the spillover elasticity, and steady-state matching remains distinct from transition implementation.

**Related literature** This paper contributes to three strands of literature in macroeconomics. The first studies global equilibrium dynamics and solution methods under aggregate uncertainty, where characterizing equilibrium dynamics with aggregate shocks has motivated extensive methodological development. [Marcet \(1988\)](#) and [Den Haan and Marcet \(1990\)](#) pioneered simulation-based solution methods that parameterize the expectation function under rational expectations; [Krusell and Smith \(1998\)](#) instead posited boundedly rational forecasters and found that a simple linear forecasting rule in aggregate capital achieves remarkable accuracy ( $R^2 > 0.9999$ ) despite the infinite-dimensional state space. Subsequent work sharpened accuracy and efficiency through perturbation and projection methods ([Re-](#)

iter, 2010; Winberry, 2018; Bhandari et al., 2023), simulation and aggregation approaches (Den Haan and Rendahl, 2010; Judd et al., 2011), sequence-space methods (Auclert et al., 2021; Azinovic-Yang and Žemlička, 2025), and deep learning (Maliar et al., 2021; Azinovic et al., 2022; Fernández-Villaverde et al., 2023; Han et al., 2026). Rather than reducing the state, Childers (2018) linearizes in function space, treating the entire distribution as the state, with consistency guarantees. On the theoretical side, Miao (2006) and Cao (2020) establish existence of competitive and recursive equilibria in this class; Proehl (2025b) gives conditions for existence and uniqueness under aggregate and idiosyncratic risk, and Proehl (2025a) develops a companion global solution method. The framework here is complementary, taking the recursive equilibrium as given and characterizing the geometry of its dynamics.

The conditional steady state links this framework to Markov-switching DSGE. Bianchi (2016) defines it as the value a state approaches while a regime persists; Chung et al. (2007) study the same object as the regime-dependent steady state, anchoring regime-dependent decision rules.<sup>1</sup> Here the point is instead the limit of an orbit, the closure of the frozen-regime trajectory: the steady state is where the path ends rather than where the analysis begins. Proposition 4 orders it against its perfect-foresight counterpart, and monotonicity along the path turns it into a coordinate.

The sequence-space Jacobian method of Auclert et al. (2021) reduces dimension by linearizing around a steady state; conditional saddle paths instead operate globally, exploiting monotonicity of an aggregate coordinate along the frozen-regime path. The two coincide to first order near a conditional steady state, but away from it the saddle bends, capturing state-dependent dynamics that first-order methods cannot, such as the regime-dependent slope asymmetries frictions generate (Section 3).

This paper differs from these computational contributions by providing geometric foundations for *why* dimension reduction works: aggregate capital’s within-spell sufficiency follows from nullcline invariance and monotonicity rather than numerical coincidence. This also has computational implications, since along each conditional saddle aggregate capital provides a theoretically justified metric for the repeated transition method of Lee (2026a).<sup>2</sup>

---

<sup>1</sup>Foerster et al. (2016) show that no single point is stationary under every regime and across every transition. Each  $K_A^{cs}$  is stationary under its own regime alone, and its failure to be stationary after a switch is what carries the economy across saddles.

<sup>2</sup>Freezing the aggregate state and analyzing transitions appears in several computational approaches to heterogeneous-agent models, including Bourany (2018), Section 4.3 of a working paper version of Achdou et al. (2022), and Lee (2026a). In continuous time, the master-equation literature (Achdou et al., 2022; Bilal, 2023) treats the distribution itself as the state and asks when finite-dimensional projections suffice, the rigorous

Second, this paper extends the geometric tradition of phase-diagram analyses in growth theory (Solow, 1956; Ramsey, 1928; Koopmans, 1963; Cass, 1965) to stochastic environments. Relatedly, the stable transformation path of Buera et al. (2024) extends the balanced growth path, and its turnpike property, to non-stationary growth models, defining a stable path independent of initial conditions. Conditional saddle paths extend the same classical geometry along a different dimension, from deterministic to stochastic settings, providing a visual tool for how microfounded frictions generate state-dependent dynamics (Kaplan and Violante, 2014; Vavra, 2014; Berger and Vavra, 2015; Basu and Bundick, 2017; Bloom et al., 2018; Kaplan et al., 2018; Petrosky-Nadeau et al., 2018; Rognlie et al., 2018; Baley and Blanco, 2019; Pizzinelli et al., 2020; Berger et al., 2021; Winberry, 2021; Lee, 2026b; Melcangi, 2024).

Finally, the random dynamical systems literature (Arnold, 1998; Schenk-Hoppé, 2001) focuses on history-dependent stochastic saddle paths (Yannacopoulos, 2011) that vary with the realized shock sequence.<sup>3</sup> By contrast, conditional saddle paths are history-invariant geometric objects, the direct analogue of deterministic phase diagrams.

## 2 Conditional saddle path

### 2.1 Definitions and assumptions

I consider a generic dynamic stochastic model whose recursive competitive equilibrium (RCE) is characterized by an aggregate state  $S$  and laws of motion  $(\Gamma_{\text{endo}}, \Gamma_{\text{exo}})$ :

$$S = [\Phi, A] \tag{1}$$

where  $\Phi$  is the endogenous and  $A$  the exogenous aggregate state, the latter possibly multivariate. The exogenous law  $\Gamma_{\text{exo}}$  is a Markov chain; for the illustration I take a two-state chain with  $A \in \{B, G\}$  and  $\Gamma_{\text{exo}}(A'|A) > 0 \forall A', A$ .

The distributional state space  $\mathcal{X}$  has elements  $\Phi$  summarizing the cross-sectional distribution of payoff-relevant idiosyncratic household states (e.g., assets and productivity types)

---

mathematical home of this question. Conditional saddle paths formalize the underlying geometric objects, and the sufficiency results explain when dimension reduction is exact along frozen-regime paths.

<sup>3</sup>In Random Dynamical Systems language, the conditional steady state is a deterministic realization of the random fixed point (Schenk-Hoppé and Schmalfuß, 2001), and the conditional saddle path is the invariant manifold of the frozen-regime dynamics. The synchronization theorem of Section 4.1 (Theorem 2) uses exactly this pullback random-fixed-point construction, recovering the current state as the long-history limit of trajectories started ever further in the past under the same regime history.

plus any endogenous objects needed to evaluate decision rules. Formally,  $\mathcal{X}$  is a subset of a metric space of probability measures, possibly augmented by a finite-dimensional vector of aggregate variables. Its metric  $d$  treats two states as close when the averages they assign to any well-behaved function of the individual state are close, with the finite-dimensional components compared in the Euclidean norm.<sup>4</sup> Throughout,  $\Gamma_{\text{endo}}(\cdot, A) : \mathcal{X} \rightarrow \mathcal{X}$  maps the current distributional state to next period's under frozen regime  $A$ .

The central object formalizes a natural thought experiment: the economy sits at  $(\Phi_0, A)$  today and the regime stays at  $A$  indefinitely. The resulting equilibrium trajectory, computed under decision rules that fully internalize future regime changes, is a frozen-regime orbit. Its closure (visited states plus their limit points) is the conditional saddle path. In aggregate phase diagrams the associated regime-specific locus is a conditional policy graph, to which the slope and curvature statements below refer.

**Definition 1** (Frozen-regime orbit and conditional saddle path).

Fix a regime  $A \in \{B, G\}$  and an initial state  $\Phi_0 \in \mathcal{X}$ . Let  $\{\Phi_t\}_{t \geq 0}$  denote the frozen-regime continuation under  $A$ , defined recursively by

$$\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A), \quad t \geq 0. \quad (2)$$

The frozen-regime orbit is

$$O(\Phi_0, A) := \{\Phi_t : t \geq 0\}. \quad (3)$$

The conditional saddle path under  $A$  from  $\Phi_0$  is defined as the closure of the frozen-regime orbit:

$$\mathcal{M}(\Phi_0, A) := \overline{O(\Phi_0, A)}. \quad (4)$$

Two projection components recur: a *coordinate*  $e : \mathcal{X} \rightarrow \mathbb{R}$ , a predetermined aggregate variable such as capital  $K$  or a knowledge stock  $H$ ; and an *aggregate object*  $g(\Phi, A)$ , an equilibrium outcome such as consumption or investment.

**Definition 2** (Conditional policy graph).

For a regime  $A$ , a coordinate  $e$ , and an aggregate object  $g$ , the conditional policy graph in the  $(e, g)$  projection is

$$\mathcal{P}_A^g := \{(e(\Phi), g(\Phi, A)) : \Phi \in \mathcal{D}_A\}, \quad (5)$$

---

<sup>4</sup> $d$  is any metric for which  $d(\Phi_n, \Phi) \rightarrow 0$  if and only if  $\int f d\Phi_n \rightarrow \int f d\Phi$  for every bounded continuous function  $f$ ; this is the standard notion of weak convergence.

where  $\mathcal{D}_A := \bigcup_{\Phi_0 \in \mathcal{X}_0} \mathcal{M}(\Phi_0, A)$  for a specified set of initial states  $\mathcal{X}_0 \subseteq \mathcal{X}$ , suppressed in the notation; the applications take  $\mathcal{X}_0$  to be the simulated ergodic support or, in Section 4.2, the sampled basin.

When  $e$  is injective (one-to-one) on a given conditional saddle path, the restriction of  $\mathcal{P}_A^g$  to that path is the graph of a single-valued function of  $e$ . Over the pooled domain  $\mathcal{D}_A$  the graph is in general a correspondence: distinct orbits can pass through the same  $e$  at different  $g$  (Section 4.1). When  $e(\Phi)$  is itself the full endogenous state, as in the representative-agent economies below,  $g$  depends on  $\Phi$  only through  $e$  and the pooled graph is single-valued.

Based on the conditional saddle path, I define the conditional steady state:

**Definition 3** (Conditional steady state).

Given  $(\Phi_0, A)$ , if the frozen-regime continuation  $\{\Phi_t\}_{t \geq 0}$  converges, I denote its limit by

$$\Phi^{cs}(\Phi_0, A) := \lim_{t \rightarrow \infty} \Phi_t. \quad (6)$$

Fixing  $A$  makes the law of motion deterministic on the distribution space: from  $\Phi_0$ ,  $\{\Phi_t\}_{t \geq 0}$  is generated by repeated application of  $\Gamma_{\text{endo}}(\cdot, A)$ . Thus  $\mathcal{M}(\Phi_0, A)$  is the natural analogue of the saddle arm in deterministic saddle-path analysis, isolating the equilibrium states (and their limit points) visited along the convergence dynamics under regime  $A$ .<sup>5</sup>

For the conditional saddle to serve as a useful geometric object, a one-dimensional curve traversed during each frozen-regime spell, the dynamics must converge to a unique limit and vary smoothly. The following assumption collects these conditions.

**Assumption 1** (Regularity of conditional saddle paths).

Fix a regime  $A \in \{B, G\}$  and an initial state  $\Phi_0 \in \mathcal{X}$ . Let  $\{\Phi_t\}_{t \geq 0}$  be defined by  $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$ .

- (i) (Unique existence) On  $\mathcal{M}(\Phi_0, A)$ , the conditional steady state  $\Phi^{cs}(\Phi_0, A)$  uniquely exists.
- (ii) (Continuity) The map  $\Gamma_{\text{endo}}(\cdot, A) : \mathcal{X} \rightarrow \mathcal{X}$  is continuous. Moreover, the aggregate variables (e.g., aggregate capital  $K$  and consumption  $C$ ) vary continuously in  $\Phi$  when restricted to  $\mathcal{M}(\Phi_0, A)$ .

---

<sup>5</sup>Strictly, in discrete time  $\mathcal{M}(\Phi_0, A) = \overline{\mathcal{O}(\Phi_0, A)}$  is an orbit closure rather than a smooth manifold, and the saddle-path label refers to its path-like role. Slope and curvature therefore refer to the policy graph  $\mathcal{P}_A^g$ . In continuous time the frozen-regime object is a one-dimensional invariant manifold.

Part (i) requires that, conditional on regime  $A$  and initial state  $\Phi_0$ , the frozen-regime continuation converge to a unique conditional steady state; it asserts neither global uniqueness across initial conditions nor the absence of multiplicity across regimes.<sup>6</sup> Part (ii) imposes continuity of the law of motion and aggregate observables on the conditional saddle, so small perturbations of the state produce small changes in aggregate outcomes. Assumption 1(i) thus restricts attention to convergent continuations, ruling out periodic orbits or divergent paths. In the RBC benchmark convergence follows from diminishing returns: the marginal product of capital is high below the conditional steady state and low above it, so capital approaches it from either side. In the heterogeneous-household economy convergence is instead a maintained condition on the equilibrium law of motion.

With this regularity the conditional saddle carries a notion of position along the orbit. Theorem 1 (Section 4.1) makes the implication exact: an aggregate variable that moves strictly one-way along the path is a coordinate for the entire conditional saddle, an exact one-dimensional representation on  $\mathcal{M}(\Phi_0, A)$ . The result is *pathwise*, orbit by orbit, regardless of the number of predetermined states; with two or more stocks the orbits fill a *surface* rather than a curve, on which a single monotone coordinate need not exist. Section 4.2 analyzes this multi-state case, where the surface also holds the paper’s planner–equilibrium geometry.

## 2.2 Representative-agent economy: A canonical RBC

A representative household with log utility maximizes lifetime utility under stochastic aggregate TFP  $A_t$ , given  $(a_0, A_0)$ . I use the recursive competitive equilibrium, equivalent to the sequential equilibrium under standard regularity conditions. The recursive form is:

$$v(a; K, A) = \max_{c, a'} \log(c) + \beta \mathbb{E}v(a'; K', A') \quad (7)$$

$$c + a' = a(1 + r(K, A)) + w(K, A) \quad (8)$$

$$a' \geq -\bar{a} \quad (9)$$

$$K' = \Gamma_{\text{endo}}(K, A), \quad A' \sim \Gamma_{\text{exo}}(A'|A), \quad (10)$$

where  $v$  is the value function,  $a$  beginning-of-period wealth,  $K$  aggregate capital,  $A$  aggregate TFP, and  $\Gamma_{\text{endo}}$  the law of motion for  $K$ . TFP  $A$  follows a Markov-switching process between

---

<sup>6</sup>Proehl (2025b) establishes global existence and uniqueness of the recursive equilibrium under explicit conditions on primitives, whereas Assumption 1(i) allows global multiplicity as in Walsh and Young (2024); conditional on a given initial state and fixed regime, the equilibrium path is assumed uniquely determined.



levels  $B$  and  $G$ , with  $G > B$ :

$$\Gamma_{\text{exo}} = \begin{bmatrix} \pi_{BB} & \pi_{BG} \\ \pi_{GB} & \pi_{GG} \end{bmatrix} \quad (11)$$

Competitive factor prices from a CRS Cobb–Douglas production function are:

$$r(K, A) = A\alpha K^{\alpha-1} - \delta \quad (12)$$

$$w(K, A) = A(1 - \alpha)K^\alpha, \quad (13)$$

where capital  $K$  satisfies  $K = a$  in equilibrium. The recursive competitive equilibrium (RCE) is:

**Definition 4** (Recursive competitive equilibrium).

$(c, a', v, r, w, \Gamma_{\text{endo}})$  is a recursive competitive equilibrium if:

1. given prices  $(r, w)$ , the policies  $(c, a')$  attain the value  $v$  in the household problem;
2. prices satisfy the competitive factor-price conditions above;
3. consistency holds:

$$a'(K; K, A) = K' = \Gamma_{\text{endo}}(K, A). \quad (14)$$

Using  $\Gamma_{\text{endo}}$  from Definition 4, I define the conditional  $K$ -nullcline.

**Definition 5** (Conditional  $K$ -nullcline in RBC).

Fix a regime  $A \in \{B, G\}$ . The conditional  $K$ -nullcline is the zero-net-investment locus in the  $(K, C)$  phase space,

$$\mathcal{N}(A) := \{(K, C) \in \mathbb{R}_{++} \times \mathbb{R}_{++} : \Delta K(K, C; A) = 0\}, \quad (15)$$

where  $\Delta K(K, C; A)$  is the one-period change in capital implied by the resource constraint at  $(K, C)$  under regime  $A$ . When  $\mathcal{N}(A)$  is the graph of a function, I write  $\mathcal{N}(A) = \{(K, C) : C = C_A^{K\text{null}}(K)\}$ .

Along a strictly monotone frozen-regime continuation, the orbit meets  $\mathcal{N}(A)$  only in the limit, at the conditional steady state; the locus is thus distinct from the *realized nullcline crossings*  $\{(K_t, C_t) : t \geq 0, K_{t+1} = K_t\}$ , empty before the limit. The resource constraint  $Y = C + I$  with  $I = K_{t+1} - (1 - \delta)K_t$  gives  $\Delta K = AK^\alpha - \delta K - C$ , so  $\mathcal{N}(A)$  is an explicit function graph.

**Proposition 1** (Characterizing the conditional  $K$ -nullcline in RBC).

The conditional nullclines of aggregate capital  $K$  for  $A \in \{B, G\}$  are

$$C_A^{Knull}(K) = AK^\alpha - \delta K, \quad (16)$$

independent of any initial condition.

*Proof.* From  $\Delta K(K, C; A) = 0$  and  $\Delta K = AK^\alpha - \delta K - C$ , the nullcline is  $C = AK^\alpha - \delta K$  for each  $A$ . ■

The nullcline invokes only  $\Delta K = AK^\alpha - \delta K - C$ , so it bears no imprint of preferences, beliefs, or the distribution; the same holds for the induced *direction field*, whose net-investment sign  $\Delta K = C_A^{Knull}(K) - C$  is fixed by accounting alone. Beliefs and heterogeneity relocate the equilibrium orbit within a phase portrait whose boundary and sign structure they cannot move. This is the common root of belief-invariance under aggregate uncertainty (Proposition 3), underlying the monotonicity diagnostic of Appendix G,<sup>7</sup> and distribution-invariance in the heterogeneous-household model (Proposition 7), underlying within-spell dimension reduction.

In discrete time capital can jump a finite amount, so the projected  $(K, C)$  trajectory can develop folds or loops, pathologies absent in continuous time. I rule them out with a geometric restriction.

**Assumption 2** (No segment crossing of the conditional  $K$ -nullcline).

For each  $A \in \{B, G\}$ , along the frozen- $A$  continuation: for every pre-limit transition (every  $t \geq 0$  with  $K_{t+1} \neq K_t$ ), the open straight line segment in  $(K, C)$  connecting  $(K_t, C_t)$  and  $(K_{t+1}, C_{t+1})$  does not intersect the active regime's conditional  $K$ -nullcline  $\mathcal{N}(A)$ :

$$((K_t, C_t), (K_{t+1}, C_{t+1})) \cap \mathcal{N}(A) = \emptyset, \quad \forall t \geq 0 \text{ with } K_{t+1} \neq K_t. \quad (17)$$

Endpoints are not required to lie off the nullcline; in particular, the conditional steady state is allowed to lie on  $\mathcal{N}(A)$ , and the assumption imposes no condition at the degenerate terminal transition  $K_{t+1} = K_t$ .

Assumption 2 ensures consecutive points cannot strictly switch sides of the  $K$ -nullcline. Its companion is *unique  $K$ -stationarity*:  $K_{t+1} = K_t$  only at  $\Phi_t = \Phi^{cs}(\Phi_0, A)$ , which in the representative-agent model follows from Assumption 1(i) ( $K$  is Markov under the frozen regime) and is otherwise maintained as a pathwise hypothesis. Economically, Assumption 2

---

<sup>7</sup>All lettered appendices are collected in the Online Appendix.

rules out single-period overshooting so severe that capital jumps past its stationary level and reverses, the discrete-time counterpart of the orbit meeting  $\mathcal{N}(A)$  only in the limit.<sup>8</sup>

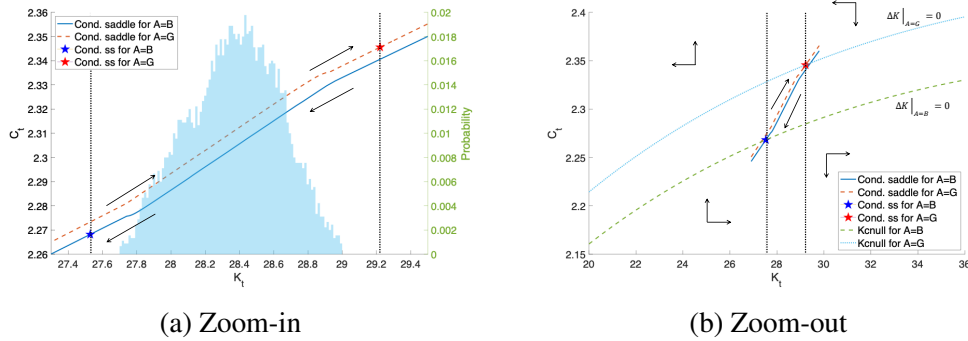
Although stated for the RBC benchmark, Assumption 2 is central to the heterogeneous-household economy (Section 4.1). It can be verified from primitives without solving the full stochastic equilibrium, conditional on a simulated frozen-regime continuation (or certified envelope bounds along it). Appendices G and E develop two complementary strategies.

**Proposition 2** (*K-monotonicity in RBC*).

Fix  $A \in \{B, G\}$ . Under Assumptions 1 and 2, together with unique *K*-stationarity ( $K_{t+1} = K_t$  only at  $\Phi_t = \Phi^{cs}(\Phi_0, A)$  along the frozen-*A* continuation), aggregate capital converges to  $K_A^{cs}$  strictly monotonically.

*Proof.* Fix  $A$  and let  $D_t := C_t - C_A^{Knull}(K_t)$  denote the nullcline gap. The resource constraint gives  $\Delta K_t = -D_t$ . By Assumption 2, no transition strictly switches sides of the nullcline:  $D_t D_{t+1} \geq 0$  for every transition  $t$  with  $K_{t+1} \neq K_t$ , and  $D_t \neq 0$  on any such transition. Unique *K*-stationarity confines  $D_{t+1} = 0$  to terminal arrival at  $\Phi_A^{cs}$ , so prior to convergence  $\{\Delta K_t\}$  has constant nonzero sign. Unique *K*-stationarity rules out nonterminal flat steps ( $\Delta K_t = 0$  at any  $\Phi_t \neq \Phi_A^{cs}$ ), so every pre-convergence transition is nondegenerate and  $\{K_t\}$  is strictly monotone. Convergence  $K_t \rightarrow K_A^{cs}$  follows from Assumption 1. ■

Figure 1: Conditional saddle paths and the phase diagram



*Notes:* Conditional *K*-nullclines ( $\Delta K = 0$ ) partition the  $(K, C)$  plane. Direction arrows show frozen-regime convergence along each conditional saddle. Panel (a) zooms into the neighborhood of the conditional steady states shown in panel (b). The histogram plots the time-series distribution of aggregate capital.

**Unique *K*-stationarity as a no-plateau condition** Unique *K*-stationarity rules out *plateau equilibria*: states off the conditional steady state at which aggregate capital stops moving

<sup>8</sup>For related geometric restrictions in dynamic optimization, see Reddy et al. (2020) and, on threshold (Skiba) phenomena, Skiba (1978), Dechert and Nishimura (1983), and Wagener (2003).

( $\Delta K_t = 0$ ) while the distribution keeps evolving, excluding the degenerate fixed points of  $K$  that arise with poverty traps, occasionally binding constraints, or policy kinks. In the representative-agent RBC,  $K$  is Markov under the frozen regime, so the condition reduces to uniqueness of the conditional steady state (Assumption 1(i)); in the heterogeneous-agent economy it holds locally via the stable eigenvalue  $\lambda_A^s \in (0, 1)$  of the linearized frozen-regime dynamics (Proposition 10). A globally certified version is more delicate: the bounded price-income MPC of Assumption 3 (Appendix E) constrains the off-steady-state dynamics, but I treat unique  $K$ -stationarity as a maintained pathwise condition, checkable along any simulated frozen-regime continuation.

In the representative-agent benchmark these restrictions are not free-standing assumptions but consequences of the scalar concave structure, given two checkable inputs verified in the computed equilibrium: a curvature-margin bound and, for stochastic beliefs, a single-crossing condition on the conditional Euler residual. Since capital is the only state, the frozen- $A$  continuation iterates a single policy  $g_A(K)$ , and Appendix J derives each restriction from a primitive: strict concavity delivers the unique conditional steady state, strictly monotone convergence, and, via the curvature-margin bound, no segment crossing, a bound standard calibrations satisfy by orders of magnitude; feasibility bounds capital; transversality enters only in selecting the stable policy. The conditions are stated as model assumptions because in the heterogeneous-agent economy of Section 4.1, the iterated state is the whole distribution, not scalar capital. Next period's  $K$  then depends on the distribution rather than on  $K$  alone. Monotone convergence requires a real stable eigenvalue of the linearized distributional dynamics, and primitives do not pin one down.

Figure 1 depicts the calibrated RBC model's conditional saddle paths in the  $(K, C)$  plane under  $A \in \{B, G\}$ , solved globally via the repeated transition method of Lee (2026a).<sup>9</sup> Panel (a) zooms into the conditional steady states with a histogram of simulated  $K_t$ ; panel (b) shows the global partition with conditional  $K$ -nullclines and direction arrows. *All* the possible dynamics of the equilibrium run along the two conditional saddles, over the capital interval bounded by their conditional steady states: within a spell the economy converges along the active saddle, and regime switches move it across; every equilibrium path, factual or counterfactual, is an alternation of these two motions, so the entire ergodic distribution is nested inside this interval.

---

<sup>9</sup>For each regime  $A$ , I hold productivity fixed for 2,500 periods and iterate the equilibrium law of motion from initial capital on both sides of  $K_A^{cs}$ , recovering the full conditional saddle in the phase diagram.

**MIT shock on the saddle** This decomposition characterizes unexpected transitory (MIT) shocks, whose link to first-order perturbation in heterogeneous-agent models is established by [Boppart et al. \(2018\)](#). For shocks representable as displacements of the predetermined state under unchanged decision rules, a shift along the conditional saddle replicates the impact, and all subsequent dynamics are along-path convergence. The slope and curvature of the conditional policy graph at the displacement point thus determine propagation, including persistence, amplification, and state dependence.

**Generalized transition function** Because conditional saddle paths capture all equilibrium transitions, along and across saddles, they support generalized transition function analysis ([Lee, 2026a](#)), encompassing generalized impulse response functions ([Koop et al., 1996](#); [Andreasen et al., 2018](#)). Any shock sequence is traced in the phase diagram as along-saddle and across-saddle movements, not confined to local dynamics around a steady state.

Figure 2 shows equilibrium dynamics in the phase diagram (panel (a)) and time domain (panels (b) and (c)) over a 20-quarter subsample. Negative TFP shocks produce downward jumps in consumption across conditional saddles, followed by endogenous decline along them; positive shocks reverse this. The phase diagram captures the dynamics in a single figure and supports counterfactual analysis. For the GIRF, one can replace the two-state

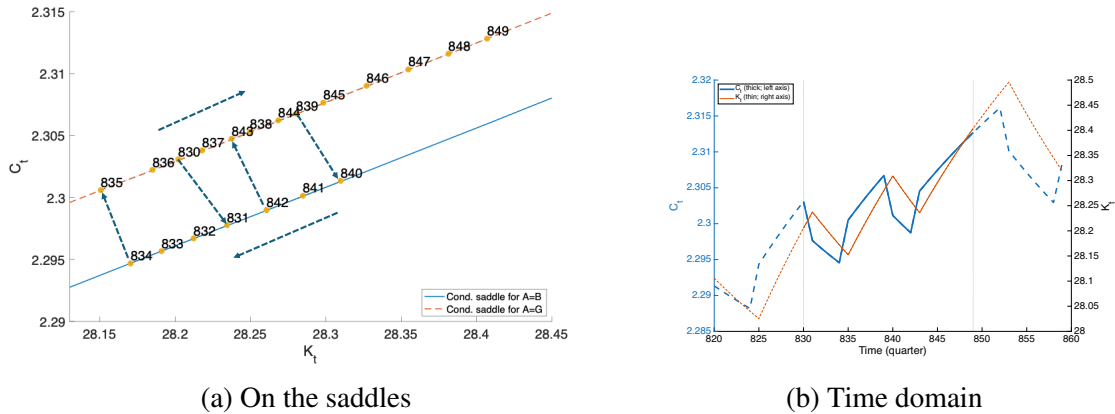


Figure 2: Equilibrium dynamics: saddle vs. time domain

*Notes:* Stochastic equilibrium dynamics on the conditional saddle paths (panel (a)) and in the time domain (panel (b),  $C_t$  thick on the left axis,  $K_t$  thin on the right; dashed segments lie outside the window). A 20-quarter window from the simulated path (quarters 830–849).

chain with a finely discretized AR(1) for log productivity,  $\ln A' = \rho \ln A + \epsilon$ ,  $\epsilon \sim N(0, \sigma^2)$ . There is then a continuum of frozen- $A$  conditional saddles indexed by  $A$ , and the GIRF path moves both along a given saddle (endogenous propagation) and across the family as  $A$

evolves.

Following a positive TFP shock, consumption jumps upward across conditional saddles on impact; thereafter, along- and across-saddle forces jointly generate a bow-shaped trajectory in the phase diagram. Capital is hump-shaped as the shock mean-reverts, and consumption first rises then falls as TFP crosses the threshold where along-saddle adjustment reverses direction.

**Comparison with the perfect-foresight saddles** In the *perfect-foresight* (PF) economy, agents believe the current regime persists forever, so the PF steady states solve  $1 = \beta(1 - \delta + \alpha A(K_A^{pf})^{\alpha-1})$ . Along the *conditional saddle* (CS), the regime is fixed ex post but agents correctly anticipate switches under  $\Gamma_{\text{exo}}$ , so the conditional steady states  $K_A^{cs}$  are pinned by probability-weighted Euler equations.<sup>10</sup> The PF and CS economies share the same  $K$ -nullcline (Proposition 3) but differ in consumption dynamics.

**Proposition 3** ( $K$ -nullcline invariance over beliefs).

*Conditional  $K$ -nullclines are identical between the RBC model with aggregate uncertainty and the perfect-foresight counterpart.*

*Proof.*

From the stationary condition  $\delta K = I$  and the aggregate resource constraint  $Y = C + I$ , equation (16) is immediate. Neither condition involves expectations or belief structures, so the nullcline is identical whether or not there is aggregate uncertainty. ■

PF–CS differences thus come entirely from consumption dynamics, not the feasibility locus; Proposition 4 formalizes this via steady-state orderings. This invariance is an accounting identity independent of beliefs and, in Section 4.1, of *who* saves, on which both the dimension-reduction result and the monotonicity diagnostic rest.

**Proposition 4** (Aggregate uncertainty and the conditional steady states).

Let  $R(A, K) := 1 - \delta + \alpha AK^{\alpha-1}$  denote the gross return and  $c(K, A')$  the stochastic-equilibrium consumption function. Suppose that (i) at each regime's PF steady state, the consumption response to a regime switch dominates the return response (jump dominance),

$$\frac{c(K_B^{pf}, G)}{c_B^{pf}} > \frac{R(G, K_B^{pf})}{R(B, K_B^{pf})}, \quad \frac{c(K_G^{pf}, B)}{c_G^{pf}} < \frac{R(B, K_G^{pf})}{R(G, K_G^{pf})}; \quad (18)$$

<sup>10</sup>Specifically,  $\beta\pi_{AA}(1 + \alpha A(K_A^{cs})^{\alpha-1} - \delta) + \beta\pi_{AA'}(c(K_A^{cs}, A')/c_A^{cs})^{-1}(1 + \alpha A'(K_A^{cs})^{\alpha-1} - \delta) = 1$  for each  $A \in \{B, G\}$ , where  $A' \neq A$ .

and (ii) each conditional Euler residual  $F_A$ , the residual of the conditional steady-state Euler equation viewed as a function of  $K$  (Appendix A), is strictly positive below  $K_A^{cs}$  and strictly negative above it. Then

$$K_B^{cs} < K_B^{pf} < K_G^{pf} < K_G^{cs}, \quad c_B^{cs} < c_B^{pf} < c_G^{pf} < c_G^{cs}. \quad (19)$$

*Proof.*

See Appendix A. ■

Both hypotheses are explicit and directly checkable. Upon a persistent regime switch, consumption moves roughly one-for-one while the gross return moves only through its marginal-product term, so condition (i) fails only in degenerate cases. Under full depreciation ( $\delta = 1$ ) the decision rules collapse to the belief-independent form of [Brock and Mirman \(1972\)](#), condition (i) holds with equality, and the two sets of steady states coincide (Appendix A); away from that corner the orderings hold in the computed equilibrium.

The two families are nested in opposite directions. The steady states nest one way: the PF pair lies inside the CS pair. Anticipated regime reversion tilts each conditional Euler equation at first order, and households (dis-)save ahead of the switch in the spirit of buffer-stock behavior ([Deaton, 1991](#); [Carroll, 1997](#)), pushing the conditional steady states apart. The *paths* nest the other way: at any common  $K$ , pricing the switch pulls each regime's consumption toward the cross-regime average, so the CS saddles run between the PF saddles. The two nestings are one mechanism: compressing consumption toward the average is (dis-)saving, and accumulating it carries the resting points apart.

Panel (a) of Figure 3 shades the path gap: one-signed regime by regime, and equal to the net-investment wedge state by state.

Panel (a) reads like the textbook surplus diagram, with the state in place of quantity: the vertical distance between the two curves is a consumption flow, and the area accumulates it across states. The  $B$  region is a consumption surplus against the perfect-foresight benchmark, financed one-for-one by slower accumulation; the  $G$  region is the consumption loss, matched by faster accumulation. Panel (b) prices the process: each conditional value saddle runs parallel to its perfect-foresight counterpart, a one-signed gap of +1.26 percent of consumption under  $B$  and −1.23 under  $G$ . In consumption-equivalent units ([Lucas, 1987](#)), this is the state-conditional welfare value of anticipated regime reversion relative to regime permanence: positive where reversion promises recovery, negative where it threatens relapse.<sup>11</sup> The near-cancellation across the two shaded gaps is geometric: with symmetric

<sup>11</sup>The absorbing benchmark bundles the direct effect of future switches with the saving response they induce.

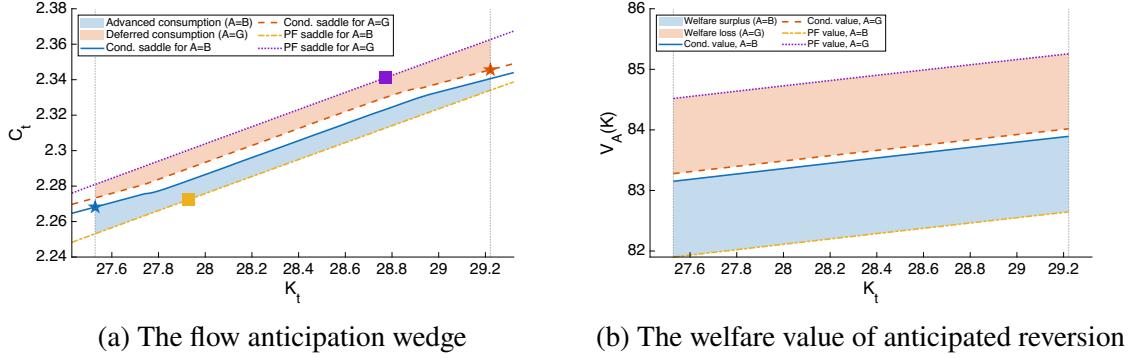


Figure 3: Policy and welfare effects of anticipated regime reversion

*Notes:* Panel (a) shades  $C_A^{cs}(K) - C_A^{pf}(K)$  on  $[K_B^{cs}, K_G^{cs}]$  (dotted verticals), +0.47 and -0.48 percent of consumption at the PF steady states; stars mark conditional and squares PF steady states. Panel (b): value saddles under the switching law against the absorbing (regime-permanence) benchmark; the gap is +1.26 and -1.23 percent in consumption equivalents.

primitives their stationary-weighted asymmetry is second order, while skewed regimes or asymmetric frictions would make it first order and visible. The flow gap varies across states; its welfare counterpart hardly does: by the envelope theorem, its consumption-equivalent slope is the panel-(a) lens scaled by  $1 - \beta$ .

**Conditional boundary condition** The recursive formulation imposes a stochastic analogue of the transversality condition, necessary in expectation (Kamihigashi, 2003, 2005). Its conditional counterpart is  $\lim_{t \rightarrow \infty} \beta^t u'(c_t) K_t = 0$  along each frozen-regime saddle. The conditional condition, holding along every history, delivers the expectation condition. The converse fails, because the frozen branch has probability zero. The conditional condition nonetheless holds in equilibrium, since the frozen continuation iterates the recursively selected policy. Appendix J establishes this from strict concavity of the one-sector problem rather than from a terminal condition.

**Economies without conditional saddle paths** A conditional saddle requires a predetermined endogenous state. Without one, the economy jumps each period to the allocation implied by the current exogenous state, so the frozen-A conditional “saddle” collapses

---

Evaluating the CS and PF policies under the same switching law isolates the welfare gain from anticipatory adjustment: positive in both regimes, approximately 0.0011 percent of permanent consumption, an optimality check on the computation. The level differences in panel (b) thus primarily reflect the welfare value of regime reversion itself.



to a singleton; I call such an economy *endogenously stateless*.<sup>12</sup> Models with predetermined states (capital, habits, interest-rate smoothing, or distributional states in HANK) are endogenously stateful, and conditional saddle paths exist under mild regularity conditions.

### 3 State dependence

Responses to the same impulse vary with the business-cycle phase: unemployment dynamics are asymmetric (Neftci, 1984), fiscal multipliers are larger in recessions (Auerbach and Gorodnichenko, 2012), and monetary policy is weaker in recessions (Tenreyro and Thwaites, 2016). I analyze this state dependence through conditional saddle paths. In any stochastic dynamic model admitting such paths, an aggregate variable's response to a shock is a shift across conditional policy graphs. The benchmark is *state independence*: the impact gap between graphs is constant in the endogenous coordinate; departures show up locally as slope differences, the first-order diagnostic.

**Lemma 1** (Vertical-separation benchmark).

*Let  $I$  be an interval over which the regime-specific conditional policy graphs  $\mathcal{P}_{A_j}^g = \{(e, g_j(e)) : e \in I\}$ ,  $j \in \{0, 1\}$ , are single-valued. Define the impact gap between the two graphs as  $\Delta_g(e; A_1, A_0) := g_1(e) - g_0(e)$ . Then conditions (i) and (ii) below are equivalent; if  $g_0, g_1$  are differentiable on  $I$ , all three are equivalent:*

- (i)  $\Delta_g$  is constant on  $I$ ;
- (ii)  $\mathcal{P}_{A_1}^g$  is a vertical translation of  $\mathcal{P}_{A_0}^g$ ;
- (iii) the slope gap  $g'_1 - g'_0$  vanishes on  $I$ .

*Proof.* The equivalence of (i) and (ii) is definitional. (i) implies (iii) by differentiating the constant gap. (iii) implies (i) by the mean value theorem:  $\Delta'_g = g'_1 - g'_0 = 0$  on the interval  $I$  makes  $\Delta_g$  constant. ■

Lemma 1 fixes the benchmark. Proposition 5 makes a substantive claim: the slope gap reflects primitives. Exact state independence is a knife-edge: any differential tilt across conditional policy graphs generates state-dependent shock responses. The degree of state dependence depends on the friction behind the tilt, whether real (Bachmann et al., 2013;

<sup>12</sup>The textbook three-equation New Keynesian model under determinacy is endogenously stateless:  $(x_t, \pi_t, i_t) = \Psi s_t$  for exogenous state  $s_t := (r_t^n, u_t, v_t)$ , so freezing  $s_t$  produces no nontrivial dynamics.

Winberry, 2021; Lee, 2026a), financial (Melcangi, 2024), or labor market (Petrosky-Nadeau et al., 2018; Pizzinelli et al., 2020). I measure the tilt locally: each regime's saddle slope at its own conditional steady state, compared across regimes as a ratio. In the log-linearized frozen-regime RBC dynamics near  $K_A^{cs}$ , let  $R_A^{cs} := 1 + f'(K_A^{cs}; A) - \delta$  denote the gross return at the conditional steady state,  $\lambda_A^s \in (0, 1)$  the stable eigenvalue, and  $\psi_A := dC/dK|_{\text{saddle}, A}$  the level saddle slope at  $K_A^{cs}$ .

**Proposition 5** (Slope ratio as a structural diagnostic).

*The level saddle slope admits the closed form*

$$\psi_A = R_A^{cs} - \lambda_A^s, \quad (20)$$

*and hence the cross-regime slope ratio is*

$$\mathcal{R} := \frac{\psi_B}{\psi_G} = \frac{R_B^{cs} - \lambda_B^s}{R_G^{cs} - \lambda_G^s}. \quad (21)$$

*Proof.* Log-linearize the frozen-regime budget constraint  $K_{t+1} = f(K_t; A) + (1 - \delta)K_t - C_t$  around  $(K_A^{cs}, C_A^{cs})$ :

$$\hat{k}_{t+1} = R_A^{cs} \hat{k}_t - \frac{C_A^{cs}}{K_A^{cs}} \hat{c}_t. \quad (22)$$

On the stable subspace of the linearized dynamics, the span of the stable eigenvector, the deviations satisfy  $\hat{c}_t = \psi_A^{\log} \hat{k}_t$  for some log slope  $\psi_A^{\log}$ . Substituting into (22) yields  $\hat{k}_{t+1} = [R_A^{cs} - \frac{C_A^{cs}}{K_A^{cs}} \psi_A^{\log}] \hat{k}_t$ , which by definition equals  $\lambda_A^s \hat{k}_t$ . Solving,  $\psi_A^{\log} = (K_A^{cs}/C_A^{cs})(R_A^{cs} - \lambda_A^s)$ . Converting to level units via  $dC/dK|_{K^{cs}} = (C_A^{cs}/K_A^{cs}) \psi_A^{\log}$  gives (20), and (21) follows by taking the ratio.

Under perfect-foresight expectations, the steady-state Euler condition gives  $R_A^{cs} = 1/\beta$  and the formula reduces to  $\psi_A = 1/\beta - \lambda_A^s$ . Under stochastic expectations over regime transitions,  $R_A^{cs}$  is pinned down by  $1 = \beta \mathbb{E}_A[(C_A^{cs}/C(\Phi_A^{cs}, A'))^\sigma (1 + f'(K_A^{cs}; A') - \delta)]$  (the return evaluated at tomorrow's capital  $K_A^{cs}$  under the realized regime  $A'$ ) and generally differs from  $1/\beta$ . The formula remains valid with  $R_A^{cs}$  evaluated at the (stochastic) conditional steady state. ■

Proposition 5 gives  $\mathcal{R}$  a structural interpretation: each regime's saddle slope is the gap between the gross return on capital at its conditional steady state and the stable eigenvalue. Once the conditional steady state is located within the stochastic equilibrium, both follow from a local linearization of the frozen-regime dynamics at that point. One reads off  $(R_A^{cs}, \lambda_A^s)$  and applies (21). Local departures from  $\mathcal{R} = 1$  indicate state-dependent impact responses. With  $\mathcal{R} > 1$  the bad-regime saddle is the steeper, so the gap between the

graphs narrows in  $K$  and the same shock moves consumption more from a low capital stock;  $\mathcal{R} < 1$  reverses the pattern.  $\mathcal{R}$  is a model-implied moment identifiable in principle from state-dependent local projections (Auerbach and Gorodnichenko, 2012; Tenreyro and Thwaites, 2016).

Since  $\mathcal{M}(\Phi_0, A)$  is an orbit closure, the slope, curvature, and tangent language of Propositions 5 and 6, of Section 4.1, and of Proposition 10 refers to its local  $C^1$  parametrization near  $K_A^{cs}$ , which exists under the regularity conditions of Proposition 10 (Appendix D).

**An asymmetric adjustment cost** Expanding the capital stock involves installation lags, ramp-up, and congestion, while contracting it largely means letting capital depreciate. If expanding is the more costly margin, the two regimes' policy graphs need not be parallel, and once they are not, the same shock moves the economy by different amounts depending on where it starts. The investment literature points the same way. Bachmann et al. (2013) document stronger aggregate investment responsiveness in expansions, and Winberry (2021) generates procyclical responsiveness through the extensive margin.

The benchmark of Section 2.2 accommodates this with a single change, an adjustment cost whose curvature differs on the two sides of zero net investment:  $C(K', K) = \frac{\tilde{\mu}}{2}((K' - K)/K)^2 K$ , where  $\tilde{\mu} = \mu_+$  if  $K' - K \geq 0$  and  $\tilde{\mu} = \mu_-$  otherwise. Its two summary parameters are the mean curvature  $\bar{\mu} := (\mu_+ + \mu_-)/2$  and the asymmetry ratio  $r := \mu_+/\mu_-$ . Two features of this specification matter. The curvature switches at net accumulation  $K' = K$  rather than at gross investment  $I = K' - (1 - \delta)K$ , so the friction bites when the economy stops adding to its capital stock, not when it stops investing altogether. And the cost is continuously differentiable at that threshold: only the second derivative jumps, so the asymmetry works through curvature and not through a kink in the cost or its slope. I read  $\mu_+ > \mu_-$  as a reduced-form expansion bottleneck, capturing installation, ramp-up, or congestion, rather than as a resale wedge.<sup>13</sup>

The threshold and orientation differ from costly reversibility: Abel and Eberly (1996) formalize it as a purchase price above the resale price, and Cooper and Haltiwanger (2006)

<sup>13</sup>The calibration sets  $\bar{\mu} = 3.05$  and  $r = 3$ , hence  $(\mu_-, \mu_+) = (1.53, 4.57)$ . The remaining annual parameters are  $\beta = 0.96$ ,  $\delta = 0.10$ , and  $\alpha = 0.33$ ;  $A \in \{0.98, 1.02\}$  has transition matrix with diagonal entries 0.90. Appendix H details the calibration: the ratio is disciplined by the downside/upside semideviation ratio of annual investment growth (PNFICA, 1967–2019) rather than the raw skewness, which is fragile to the 2009 outlier, and  $\mathcal{R}$  is reported across the robustness band  $r \in [2, 4]$ . That moment bounds  $r$  from below at about 2.3 without bounding it from above, so  $r = 3$  is a conservative choice inside the identified set rather than a structural estimate of the two curvature parameters. Relative investment volatility is untargeted, since even the frictionless limit sits within one standard error of the narrow series; the level is anchored on the investment– $q$  elasticity.

estimate a mixture of convex, nonconvex, and resale-price costs. [Zhang \(2005\)](#) uses asymmetric quadratic curvature with  $\theta^- > \theta^+$  switching at gross investment  $I = 0$ , which puts the high curvature on disinvestment rather than on expansion. The specification is a parsimonious setting for studying how an asymmetric real friction tilts policy graphs.

Two comparative statics follow: the first moves the friction pair  $(\mu_-, \mu_+)$  at fixed mean curvature, the second risk aversion (Figure 4). All same-capital comparisons use interior common-support domains. Within a given specification, define  $\mathcal{R}(K) = C'_B(K)/C'_G(K)$  at the same  $K$  over the intersection of the regimes' 5th–95th percentile policy supports,  $C_A$  being the single-valued recursive policy since  $K$  is the full endogenous state; the marginal- $q$  analogue is  $\mathcal{R}_q(K) := q_B^{K'}(K)/q_G^{K'}(K)$ , the same construction at  $g = q^K$ . Holding  $\bar{\mu} = 3.05$  fixed, the median is 1.265 under the calibrated asymmetry ( $r = 3$ ), 0.976 under symmetric curvature ( $r = 1$ ), and 0.752 under the reversed asymmetry ( $r = 1/3$ ). The friction's orientation therefore determines which regime's saddle is steeper, and reversing it moves  $\mathcal{R}$  from above one to below it. Under the calibrated tilt the separations narrow as capital rises: over the sampled range the consumption gap between the saddles falls by 14 percent and the marginal- $q$  gap by 36 percent, so how well capitalized the economy is governs how far a given switch moves it.

The upper row of Figure 4 visualizes a general-equilibrium comparative static on the geometry itself: the saddle pair for  $A = B, G$  is the object compared, and preferences are the parameter moved. For this exercise, utility extends from the log benchmark to CRRA,  $u(c) = c^{1-\gamma}/(1-\gamma)$ , with  $\gamma$  the coefficient of relative risk aversion. Every case keeps  $r = 3$ , equivalently  $(\mu_-, \mu_+) = (1.53, 4.57)$  at  $\bar{\mu} = 3.05$ , together with the same technology, shock process, and simulated shock path. The left column reports consumption saddles  $C_A(K)$ ; the right reports marginal  $q_A^K(K) = 1 + C_{K'} = 1 + \bar{\mu}(K' - K)/K$ . The top row varies the CRRA coefficient  $\gamma$ , changing relative risk aversion and the elasticity of intertemporal substitution (EIS),  $1/\gamma$ , jointly. As  $\gamma$  rises from 1 to 10, the median  $\mathcal{R}$  falls from 1.265 to 1.047. The median  $\mathcal{R}_q$  falls from 0.468 to 0.321 as both  $q$  saddles flatten. But the conditional level gap  $\mathbb{E}[q^K | G] - \mathbb{E}[q^K | B]$  rises from 0.0153 to 0.0315, and the simulated unconditional  $\text{sd}(q^K)$  rises from 0.00932 to 0.01723. Marginal  $q$  therefore becomes less sensitive to capital position within a regime but more volatile because regime switches traverse more widely separated saddles.

The bottom row of Figure 4 turns to Epstein–Zin utility, holding the EIS fixed at one while relative risk aversion rises from 1 to 10. Strikingly, the curves barely move: the median  $\mathcal{R}$  remains approximately 1.265, the median  $\mathcal{R}_q$  moves only from 0.4684 to 0.4685,

and  $\text{sd}(q^K)$  changes only from 0.00932 to 0.00933. The strong CRRA comparative statics come mainly from the accompanying fall in the EIS, not from higher risk aversion, consistent with Tallarini (2000). The anticipation gap of Figure 3 inherits this invariance.

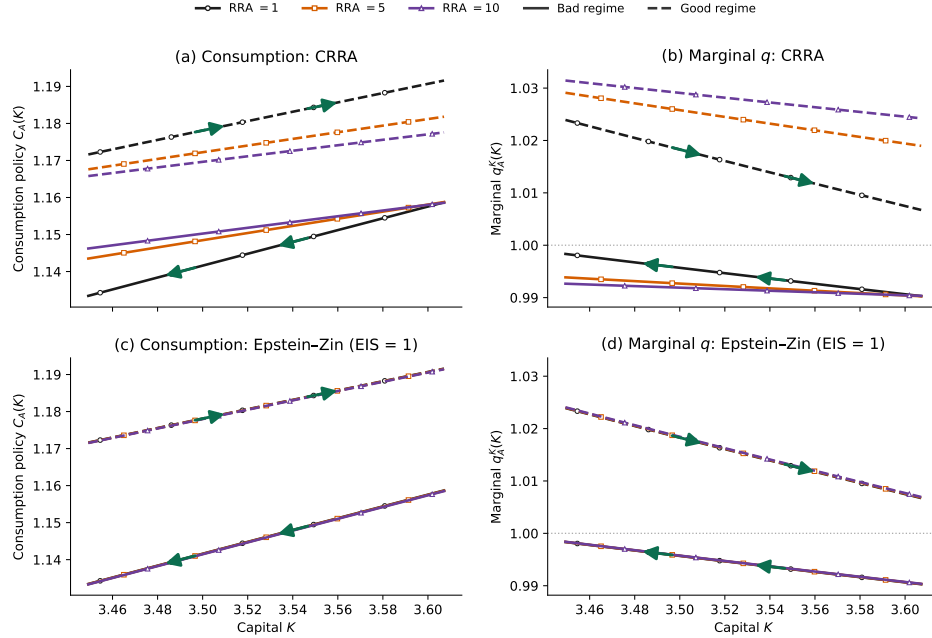


Figure 4: Consumption and marginal- $q$  saddles across preferences

*Notes:* Panels (a) and (c) plot the consumption policies  $C_A(K)$ ; panels (b) and (d) plot marginal  $q_A^K(K)$ . The top row varies risk aversion under CRRA, the bottom under Epstein–Zin at unit EIS. Colors and markers code risk aversion; solid and dashed lines the bad and good regimes. All curves are evaluated on the common support of the solved cases. Horizontal lines mark  $q^K = 1$ ; arrows give the direction of capital motion from the sign of  $q^K - 1$ . With  $q_B^K < 1 < q_G^K$  throughout, both conditional steady states lie outside the band: the economy is always in transit. Marginal  $q$  displays the assumed adjustment wedge and is not independent calibration evidence. Same-mean symmetric and sign-reversed cases are in Appendix H.

Having used the asymmetric friction to isolate slope-driven state dependence, I return to the frictionless benchmark for a distinct channel: response-shape variation from initial position along the saddle.

**Response shape in the frictionless benchmark** Although  $\mathcal{R} \approx 1$  in the benchmark RBC economy, the *shape* of the response trajectory is highly state dependent. A positive productivity shock displaces the economy across saddles; the trajectory depends on  $K_0$ 's position relative to  $K_G^{\text{cs}}$ . Far below  $K_G^{\text{cs}}$ , the marginal product of capital is high, tilting the intertemporal tradeoff toward accumulation and producing a late consumption hump (Barro and Sala-i-Martin, 1992); near  $K_G^{\text{cs}}$ , across-saddle reversion dominates and the peak arrives earlier. This channel is orthogonal to the slope asymmetry of Proposition 5: slope differences

govern *impact* magnitudes, position-dependent convergence speed governs response *shapes*. The shape does *not* require a nonlinear saddle: even in the log-linearized economy, with a straight-line saddle, peak timing varies with  $K_0$ . The mechanism is a race between the endogenous along-saddle rate  $\lambda$  and the exogenous across-saddle rate  $\rho$ , weighted by initial distance from steady state; concavity amplifies the effect but is not its source.

Near the non-conditional steady state, the Blanchard–Kahn decomposition (Blanchard and Kahn, 1980) on the saddle branch yields

$$\hat{c}_t = \underbrace{\mathcal{P}(\hat{k}_0)}_{\text{along saddle}} \lambda^t + \underbrace{Q}_{\text{across saddle}} \rho^t, \quad (23)$$

where  $\hat{c}_t$  and  $\hat{k}_0$  are log-deviations from steady state,  $\lambda \in (0, 1)$  is the stable eigenvalue of the capital dynamics, and  $\rho \in (0, 1)$  is the TFP persistence. The shock size  $\varepsilon > 0$  enters both terms. Initial capital enters only the along-saddle coefficient  $\mathcal{P}(\hat{k}_0) = \psi_k [\hat{k}_0 + \theta\varepsilon/(\lambda - \rho)]$ , where  $\psi_k > 0$  is the saddle slope and  $\theta > 0$  the capital-TFP elasticity; the across-saddle coefficient  $Q = [\psi_a - \psi_k\theta/(\lambda - \rho)]\varepsilon$  is independent of  $\hat{k}_0$ .<sup>14</sup> The hump arises from the tension between two decaying exponentials: the  $\lambda^t$  term captures along-saddle propagation, the  $\rho^t$  term direct TFP reversion, and the peak occurs where they balance. Lower initial capital reduces  $\mathcal{P}(\hat{k}_0)$ , weakening the slowly decaying along-saddle term, so the balance tips later, exactly the comparative static  $\partial t^*/\partial \hat{k}_0 < 0$  below.

**Proposition 6** (State-dependent peak timing).

Suppose  $\lambda > \rho$ ,  $\mathcal{P}(\hat{k}_0) > 0$ , and the consumption response (23) admits a peak at  $t^* > 0$ . Then

$$\frac{\partial t^*}{\partial \hat{k}_0} = \frac{-\psi_k}{\mathcal{P}(\hat{k}_0) \cdot \ln(\lambda/\rho)} < 0, \quad (24)$$

where  $\hat{k}_0$  is the log-deviation of the initial capital stock from steady state.

*Proof.*

See Appendix B. ■

The consumption peak arrives strictly later when initial capital is farther below the steady state. The hypotheses are mild:  $\lambda > \rho$  holds under standard quarterly RBC calibrations ( $\lambda \approx 0.96$ ,  $\rho \leq 0.95$ ), and  $\mathcal{P}(\hat{k}_0) > 0$  requires only  $\hat{k}_0 > -\theta\varepsilon/(\lambda - \rho)$ , which the ergodic

<sup>14</sup>This  $\lambda$  (at the non-conditional steady state) and the regime-specific  $\lambda_A^s$  of Proposition 5 (at the conditional steady state) coincide to first order in the frictionless RBC ( $\lambda_B^s = \lambda_G^s = \lambda$ , so  $\mathcal{R} \approx 1$ ); frictions generate  $\mathcal{R} \neq 1$ . Proposition 6 then operates in the  $\mathcal{R} \approx 1$  regime, its state dependence driven by initial location  $\hat{k}_0$  rather than regime-varying slopes.

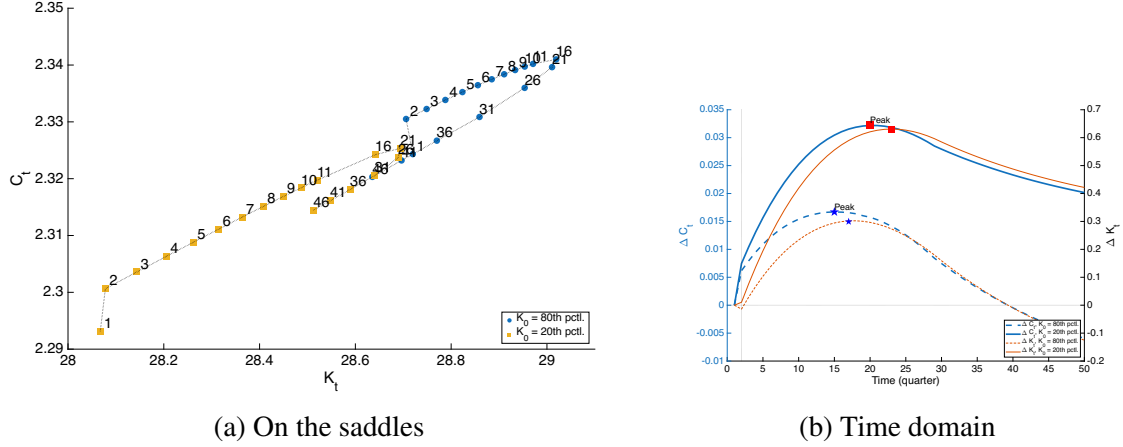


Figure 5: Shape of the impulse responses: low  $K_0$  vs. high  $K_0$

*Notes:* Impulse responses to a positive productivity shock from  $K_0$  at the 20th and 80th percentiles of the ergodic distribution. Panel (a): post-shock trajectories in  $(K, C)$  space on the conditional saddle paths; numbers indicate quarters since impact. Panel (b): time-domain responses,  $\Delta C_t$  on the left axis and  $\Delta K_t$  on the right; markers indicate the peak quarters. Sample: 50 quarters.

distribution satisfies with substantial margin. Proposition 6 is local by construction, and Figure 5 shows the mechanism operating globally. Comparing the 20th and 80th percentiles of the ergodic capital distribution, from low  $K_0$  both  $C$  and  $K$  rise longer before reversion dominates, producing a later and larger consumption peak, while from high  $K_0$  the peak arrives earlier. The same saddle geometry, shock, and preferences apply; only the inherited capital stock differs.

## 4 Applications

The two applications extend the RBC benchmark of Section 2.2 in complementary directions. The heterogeneous-household economy (Section 4.1) adds idiosyncratic income risk; monotonicity lets aggregate capital index each spell of the frozen orbit, despite its true distributional state. The investment-externality economy (Section 4.2) instead adds investment-generated aggregate knowledge: the second inherited stock turns the family of frozen paths into a *surface*, whose gap from the planner's surface identifies the distortion and optimal policy.

## 4.1 Heterogeneous households: dimension reduction

In a heterogeneous-household economy, the natural state is the entire cross-sectional wealth distribution, an infinite-dimensional object. Classical aggregation ([Gorman, 1961](#)) reduces dimensionality by restricting preferences, but these restrictions fail under uninsurable idiosyncratic risk and borrowing constraints. Yet practitioners routinely approximate the state by a single variable, aggregate capital. The conditional saddle framework explains why this works without preference restrictions by splitting the claim into two halves of very different difficulty: *within* a frozen-regime spell the distribution evolves along a one-dimensional orbit, so aggregate capital is an *exact* coordinate for it. *Across* histories, the harder half, a single capital rule works only approximately, an error this section bounds and documents directly.

I consider a continuum of ex-ante homogeneous households of unit measure, with recursive problem

$$v(a, z; \Phi, A) = \max_{c, a'} \log(c) + \beta \mathbb{E}v(a', z'; \Phi', A') \quad (25)$$

$$c + a' = a(1 + r(\Phi, A)) + w(\Phi, A)z \quad (26)$$

$$a' \geq 0 \quad (27)$$

$$\Phi' = \Gamma_{\text{endo}}(\Phi, A), \quad A' \sim \Gamma_{\text{exo}}(A'|A). \quad (28)$$

The problem differs from the RBC benchmark economy only in (i) uninsurable idiosyncratic labor productivity ([Bewley, 1986](#); [Huggett, 1993](#); [Aiyagari, 1994](#)), a Markov process  $z \sim \Gamma_z(z'|z)$ ; (ii) inclusion of the distribution  $\Phi$  in the aggregate endogenous state; and (iii) the corresponding law of motion. The RCE is defined as in [Krusell and Smith \(1998\)](#).

The model has two exogenous stochastic processes, idiosyncratic productivity and aggregate TFP, hence two layers of conditional saddle paths, individual and aggregate. I focus on the aggregate path under stochastic TFP, following [Krusell and Smith \(1998\)](#).<sup>15</sup> Consistent with Section 2.2,  $K = K(\Phi)$  and  $C = C(\Phi, A)$  denote aggregate capital and consumption.

The conditional  $K$ -nullcline of Definition 5 extends directly to the heterogeneous-household setting: net investment depends on  $\Phi$  only through the aggregates  $(K, C)$ , so the

<sup>15</sup>In the original [Krusell and Smith \(1998\)](#) model exogenous individual labor supply co-moves with aggregate TFP, and the computation follows that specification. Aggregate labor  $L_A$  is constant within a regime, so output is  $AK^\alpha L_A^{1-\alpha}$  and the frozen-regime analysis is the fixed-labor one with  $A$  replaced by the effective productivity  $AL_A^{1-\alpha}$ . I write  $A$  for that product below. With the unemployed supplying 0.25 efficiency units, regime-conditional labor is  $L_B = 0.929$  and  $L_G = 0.970$ , and the effective regime gap is 4.9 percent, 2.4 times the raw TFP gap.



locus remains an object in the aggregate phase plane.

**Definition 6** (Conditional  $K$ -nullcline in the heterogeneous-household setting).

Fix  $A \in \{B, G\}$ . The conditional  $K$ -nullcline is the zero-net-investment locus in the aggregate  $(K, C)$  plane,

$$N(A) := \{(K, C) \in \mathbb{R}_{++} \times \mathbb{R}_{++} : \Delta K(K, C; A) = 0\},$$

where  $\Delta K(K, C; A)$  is the one-period change in aggregate capital implied by the aggregate resource constraint at  $(K, C)$  under regime  $A$ . As in the representative-agent case,  $N(A)$  is distinct from the realized crossings  $\{(K_t, C_t) : K_{t+1} = K_t\}$  along any frozen-regime continuation  $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$ , which under strict  $K$ -monotonicity contain no point at all before the limit.

Krusell and Smith (1998) posits an endogenous law of motion tracking aggregate capital  $K$  rather than the full distribution  $\Phi$ :  $\log K' = b_0(A) + b_1(A) \log K$ . This embeds two distinct assumptions: (i) aggregate capital  $K$  suffices to summarize the endogenous aggregate state, and (ii)  $K$  follows a log-linear law of motion. Even the existence of a recursive equilibrium on a small endogenous state space is delicate (Kubler and Schmedders, 2003).

Since Krusell and Smith (1998), it has been an open question whether  $K$ -based forecasting works so well ( $R^2 > 0.9999$ ) by accident or through a deeper structural property.<sup>16</sup> The within-spell half of (i) is settled exactly: strict monotonicity along a single orbit, established below, makes  $K$  index the distributional state path by path, so conditioning on  $K$  loses no within-spell information. The across-history half asks why approximately the *same* rule works from every distributional state, not merely along one orbit; I return to it after the within-spell results.

By contrast, (ii) is a different kind of assumption. Even with  $K$  as the coordinate, the true law of motion for  $K$  is generally non-Markovian, because the omitted distribution still influences next period's capital, and it need not be log-linear, so any parametric rule remains an approximation. This is a question of functional form rather than coordinate choice, and the paper's subject is the latter.

The following theorem is the first step toward exact conditional sufficiency of  $K$ . If the economy converges along a one-dimensional equilibrium path and some aggregate variable moves strictly in one direction throughout, its current value identifies exactly where the economy is, since no two distinct distributional states share the same value.

---

<sup>16</sup>Den Haan (2010) shows the forecasting  $R^2$  can substantially overstate dynamic accuracy, because one-step-ahead fit masks the accumulation of errors over long simulated paths; the across-spell handoff check reported below is in the spirit of his stricter multi-period standard.

**Theorem 1** (Monotone aggregate variable as a coordinate).

Fix  $(\Phi_0, A_0)$  and  $A \in \{B, G\}$ . Let  $\{\Phi_t\}_{t \geq 0}$  be the frozen-regime continuation  $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$  and let  $\Phi^{cs} := \lim_{t \rightarrow \infty} \Phi_t$ . Let  $e : \mathcal{M}(\Phi_0, A) \rightarrow \mathbb{R}$  be an aggregate equilibrium variable. Let  $t^* \leq \infty$  be the first date at which  $\Phi_t = \Phi^{cs}$ . Suppose that the scalar sequence  $\{e(\Phi_t)\}_{t \geq 0}$  is strictly monotone for  $t < t^*$  and converges to  $e(\Phi^{cs})$ . Then  $e$  uniquely indexes states on the conditional saddle:

$$e(\Phi) = e(\Phi') \implies \Phi = \Phi' \quad \forall \Phi, \Phi' \in \mathcal{M}(\Phi_0, A). \quad (29)$$

*Proof.* Define  $\iota_A : \mathbb{N} \cup \{\infty\} \rightarrow \mathcal{M}(\Phi_0, A)$  by  $\iota_A(t) := \Phi_t$  for  $t \in \mathbb{N}$  and  $\iota_A(\infty) := \Phi^{cs}$ . By definition of  $\mathcal{M}(\Phi_0, A)$  and  $\Phi_t \rightarrow \Phi^{cs}$ ,

$$\mathcal{M}(\Phi_0, A) = \{\Phi_t : t \geq 0\} \cup \{\Phi^{cs}\}, \quad (30)$$

so  $\iota_A$  is surjective. Take first  $t^* = \infty$ . Strict monotonicity of  $\{e(\Phi_t)\}$  implies  $e(\Phi_t) \neq e(\Phi_\tau)$  for  $t \neq \tau$ , hence  $\Phi_t \neq \Phi_\tau$ , so  $\iota_A$  is injective on  $\mathbb{N}$ ; convergence to  $e(\Phi^{cs})$  gives  $e(\Phi_t) \neq e(\Phi^{cs})$  for all finite  $t$ , so  $\iota_A$  is injective on  $\mathbb{N} \cup \{\infty\}$ , hence a bijection. If instead  $t^* < \infty$ , the orbit is the finite set  $\{\Phi_0, \dots, \Phi_{t^*}\}$ , on which  $e$  takes  $t^* + 1$  distinct values by strict monotonicity up to  $t^*$ , so  $\iota_A$  restricted to  $\{0, \dots, t^*\}$  is again a bijection. For  $\Phi = \iota_A(s)$ ,  $\Phi' = \iota_A(s')$ ,  $e(\Phi) = e(\Phi')$  gives  $e(\iota_A(s)) = e(\iota_A(s'))$ ; since  $e \circ \iota_A$  is injective,  $s = s'$  and  $\Phi = \Phi'$ . Thus  $e$  is injective on  $\mathcal{M}(\Phi_0, A)$ , and an inverse map  $\varphi_A$  exists on  $e(\mathcal{M}(\Phi_0, A))$ . ■

Consequently, a unique inverse map  $\varphi_A$  from  $e(\mathcal{M}(\Phi_0, A))$  to  $\mathcal{M}(\Phi_0, A)$  satisfies  $\varphi_A(e(\Phi)) = \Phi$  on  $\mathcal{M}(\Phi_0, A)$ , and any equilibrium object restricted to  $\mathcal{M}(\Phi_0, A)$  is a single-valued function of the scalar coordinate  $e$ . The following proposition shows that the conditional  $K$ -nullclines coincide with those of the representative-agent model.

**Proposition 7** (Conditional  $K$ -nullclines of Krusell and Smith (1998)).

*The conditional  $K$ -nullclines of the heterogeneous-household model are identical to those of the representative-household model and invariant over the initial distribution  $\Phi_0$ :*

$$C_A^{Knull}(K) = AK^\alpha - \delta K, \text{ for } A \in \{B, G\}. \quad (31)$$

*Proof.*

As in Proposition 3, the stationary condition  $\delta K = I$  and the aggregate resource constraint  $Y = C + I$  immediately imply the form of the conditional nullclines. ■

As anticipated after Proposition 1, this is the distribution-invariance instance of that common root, and it is what enables dimension reduction. The boundary governing the sign

of capital accumulation, an accounting identity independent of *who* saves, does not move as the distribution evolves along the conditional saddle.

The invariance is specific to the neoclassical production structure with exogenous labor supply. Under separable CRRA preferences with wealth effects on labor, labor supply introduces a distributional channel into the resource constraint; the nullcline then acquires distributional dependence, blocking the analytical route to  $K$ -sufficiency. However, monotonicity remains verifiable in those setups: the Appendix G check certifies it along the computed frozen-regime continuation, without invoking nullcline invariance.

**Proposition 8** ( $K$  monotonicity and injectivity).

*Fix  $A \in \{B, G\}$ . Under Assumptions 1 and 2, together with unique  $K$ -stationarity along the frozen- $A$  continuation (the no-plateau condition of Section 2.2), aggregate capital converges to  $K_A^{cs}$  strictly monotonically on  $\mathcal{M}(\Phi_0, A)$ . Consequently,  $K$  is injective on  $\mathcal{M}(\Phi_0, A)$ .*

*Proof.*

By Proposition 7, the conditional  $K$ -nullcline takes the RBC form, so strict monotonicity follows the proof of Proposition 2 verbatim with  $C_A^{Knull}(K) = AK^\alpha - \delta K$ . Unique  $K$ -stationarity enters locally through the stable-eigenvalue condition and globally as a pathwise hypothesis verifiable by inspecting the simulated frozen-regime continuation. The primitive MPC condition (Appendix E, Lemma E.1) ensures no side-switching but does not rule out non-steady-state flat steps. Since  $\{K_t\}$  is strictly monotone and converges to  $K_A^{cs}$ , Theorem 1 with  $e(\Phi) = K(\Phi)$  yields  $K(\Phi) = K(\Phi') \implies \Phi = \Phi'$  for all  $\Phi, \Phi' \in \mathcal{M}(\Phi_0, A)$ . ■

Proposition 7 makes the  $K$ -nullcline *sign-determining*: the graph  $C = C_A^{Knull}(K)$  partitions the  $(K, C)$  plane into regions with  $\Delta K > 0$  and  $\Delta K < 0$  via  $\Delta K = C_A^{Knull}(K) - C$ . Assumption 2 rules out discrete-time jumps across this partition, so net investment cannot reverse sign along the conditional saddle. Adding unique  $K$ -stationarity (the no-plateau condition) to rule out non-steady-state flat steps, capital dynamics are strictly monotone and each frozen-regime spell admits an exact one-dimensional representation indexed by  $K$ . Proposition 8 makes  $K : \mathcal{M}(\Phi_0, A) \rightarrow \mathbb{R}$  one-to-one, so an inverse map  $\varphi_{\Phi_0, A}$  exists with  $\Phi = \varphi_{\Phi_0, A}(K(\Phi))$  on  $\mathcal{M}(\Phi_0, A)$ .<sup>17</sup> Any equilibrium object restricted to  $\mathcal{M}(\Phi_0, A)$  then admits an exact single-valued representation as a function of  $(K, A)$ .

Panel (a) of Figure 6 plots the computed conditional saddle paths in the  $(K, C)$  phase diagram. The model is solved globally using the repeated transition method, and the

<sup>17</sup>This is a path-specific notion of sufficiency: for a fixed  $(\Phi_0, A)$ ,  $K$  uniquely indexes states on the entire conditional saddle (globally along the transition path).

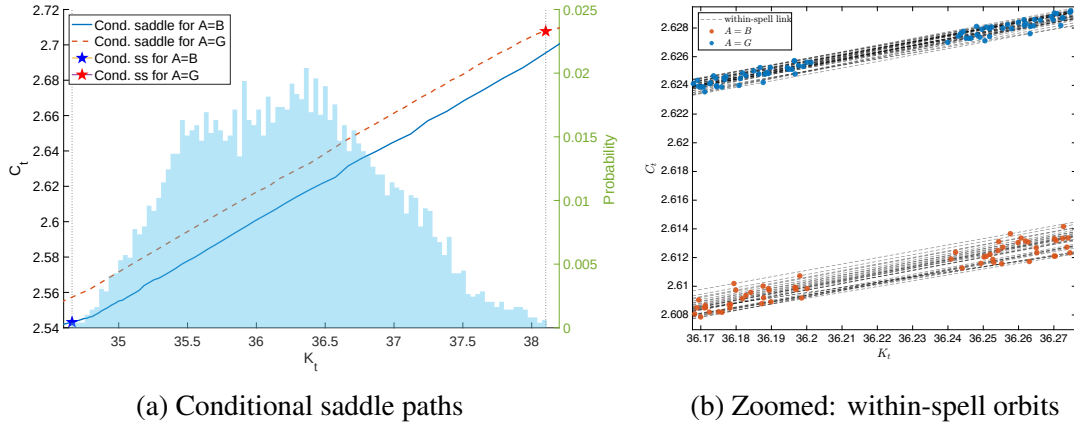


Figure 6: Conditional saddle paths in [Krusell and Smith \(1998\)](#)

*Notes:* Panel (a): conditional saddle paths for  $A = B$  (solid) and  $A = G$  (dashed); the background histogram plots the time-series distribution of aggregate capital. Panel (b): simulated ergodic  $(K_t, C_t)$  pairs in a narrow window around the median capital stock,  $K \in [36.17, 36.28]$  ( $A = B$  red,  $A = G$  blue); each dashed line fits one spell's points and extends across the window, exposing that spell's local slope.

dynamics under each frozen aggregate state are simulated for 2,500 periods. Each regime's apparent single curve in fact overlays many conditional saddle paths, one per realized spell and its inherited distribution  $\Phi_0$ . Although the true endogenous state is  $\Phi$  rather than  $K$  alone, each path is strictly monotone in  $K$ , supporting the spell-wise sufficiency result.

Panel (b) zooms into a narrow  $K$ -window and uncovers the finer structure: the ergodic band, line-like at the scale of panel (a), is “dusty”. Within each spell the consecutive  $(K_t, C_t)$  points move monotonically in  $K$ , tracing an exact one-dimensional orbit, while ergodic mixing across spells produces the band observed at any given  $K$ . The within-spell links are moreover visibly parallel: the local slope is approximately invariant to the inherited distribution, anticipating its eigenvalue anchor in [Proposition 10](#).

**From spell-wise to global sufficiency** [Proposition 8](#) is stated regime-by-regime:  $K$  uniquely indexes the conditional saddle  $\mathcal{M}(\Phi_0, A)$  under a single frozen regime. The equilibrium path concatenates such one-dimensional segments, each exactly indexed by  $(K, A)$  within its spell. The handoff between spells, however, is governed by the full distributional state  $\Phi_\tau$  at the switch date, which  $(K_\tau, A_\tau)$  need not determine: different histories can arrive at the same  $(K_\tau, A_\tau)$  carrying different distributions. Spell-wise sufficiency is thus distinct from unconditional global sufficiency, which would require the same  $K$  to pin the same  $\Phi$  regardless of path history. The following proposition bounds this across-spell discrepancy.

Observing the economy at a given  $K$  admits a “cloud” of distributions from different

prior histories; three results follow. Theorem 2 supplies the mechanism: economies forget where they started, so two sharing recent regimes end up close. Proposition 9 averages that over the number of recent regimes two histories share, which persistence determines. Proposition 10 instead supplies a rate, identifying the local contraction along the saddle with the stable eigenvalue that sets its slope.

Write  $\Gamma_A := \Gamma_{\text{endo}}(\cdot, A)$ . The maintained hypothesis is included in the following definition.

**Definition 7** (Uniform frozen-regime contraction).

*The family  $\{\Gamma_A\}_{A \in \{B, G\}}$  is a uniform frozen-regime contraction on  $\mathcal{U}$  at rate  $\lambda_c \in (0, 1)$  if  $\mathcal{U} \subseteq \mathcal{X}$  is closed in the complete metric space  $(\mathcal{X}, d)$ , forward-invariant under every  $\Gamma_A$  (each  $\Gamma_A$  maps  $\mathcal{U}$  into itself), of finite diameter  $\text{diam}(\mathcal{U}) := \sup_{\Phi, \Phi' \in \mathcal{U}} d(\Phi, \Phi')$ , and*

$$d(\Gamma_A \Phi, \Gamma_A \Phi') \leq \lambda_c d(\Phi, \Phi') \quad \text{for all } \Phi, \Phi' \in \mathcal{U} \text{ and all } A.$$

**Theorem 2** (Synchronization under frozen-regime contraction).

*Let  $\{\Gamma_A\}$  be a uniform frozen-regime contraction on  $\mathcal{U}$  at rate  $\lambda_c$  (Definition 7). Then:*

- (i) *(Random fixed point: the history-dependent analogue of a contraction's unique fixed point) For every regime history  $\omega = (\dots, A_{t-2}, A_{t-1})$ , the long-history limit (the state obtained by applying the history's maps from an ever more distant start)*

$$\Phi^*(\omega) := \lim_{T \rightarrow \infty} (\Gamma_{A_{t-1}} \circ \Gamma_{A_{t-2}} \circ \dots \circ \Gamma_{A_{t-T}})(\Phi)$$

*exists, is independent of  $\Phi \in \mathcal{U}$ , and is the unique state in  $\mathcal{U}$  consistent with  $\omega$ ; any trajectory initialized in  $\mathcal{U}$  at date  $t - T$  satisfies  $d(\Phi_t, \Phi^*(\omega)) \leq \lambda_c^T \text{diam}(\mathcal{U})$ .*

- (ii) *(Suffix synchronization) If two histories  $\omega, \omega'$  agree over their most recent  $T$  periods, then*

$$d(\Phi^*(\omega), \Phi^*(\omega')) \leq \lambda_c^T \text{diam}(\mathcal{U}).$$

*Proof.*

See Appendix C. ■

The limit in part (i) runs backward. Fix the recent regime sequence and start the economy ever earlier: because every map contracts, the starting state washes out and the recent history alone pins down where the economy is now. This is a random-fixed-point construction in the spirit of [Schenk-Hoppé and Schmalfuß \(2001\)](#), and part (ii) is its consequence for two states in the same- $K$  cloud, averaged below over the common-suffix length. For the

stationary symmetric two-state chain ( $\pi_{BB} = \pi_{GG} = \rho_A$ ), let  $q := \rho_A^2 + (1 - \rho_A)^2$  denote the probability that two independent histories agreeing at a date also agree one period earlier.<sup>18</sup>

**Proposition 9** (Across-spell approximation bound).

Let  $\{\Gamma_A\}$  be a uniform frozen-regime contraction on  $\mathcal{U}$  at rate  $\lambda_c$  (Definition 7), and let  $\Phi, \Phi' \in \mathcal{U}$  be the long-history limits of two regime histories that agree over their most recent  $T$  periods. Then

$$d(\Phi, \Phi') \leq \lambda_c^T \text{diam}(\mathcal{U}).$$

If, moreover, the two histories are drawn independently from the stationary symmetric chain and share the current regime, then  $\Pr(T = k) = (1 - q)q^k$  for  $k \geq 0$  and

$$\mathbb{E}[d(\Phi, \Phi')] \leq \frac{1 - q}{1 - q\lambda_c} \text{diam}(\mathcal{U}).$$

*Proof.*

See Appendix C. ■

The count  $T$  begins at the previous date, because the shared current regime indexes the cloud but has not yet acted on  $\Phi$ . The diameter  $\text{diam}(\mathcal{U})$  is finite under the borrowing constraint  $a' \geq 0$  and compact idiosyncratic support. The right-hand side involves only the primitive set  $\mathcal{U}$ , not the same- $K$  cloud itself, so the proposition is a genuine bound on the cloud rather than a contraction statement about it. The bound vanishes as  $\rho_A \rightarrow 1$ , since long spells make recent suffixes long. It does not vanish as  $\lambda_c \rightarrow 0$ : a pair sharing no regime history is bounded only by  $\text{diam}(\mathcal{U})$ , and that happens with probability  $1 - q$ .

**Assumed uniform contraction** The hypothesis that  $\Gamma_{\text{endo}}(\cdot, A)$  contracts uniformly on  $\mathcal{U}$  is not innocuous: the map embeds general-equilibrium price feedback, and establishing contraction-type properties for such operators is precisely the difficulty confronted by global existence and uniqueness analyses (Proehl, 2025b). Locally, Proposition 10 anchors the contraction rate to the stable eigenvalue  $\lambda_A^s \in (0, 1)$ ; globally the assumption is maintained, and the sample-based diagnostic below provides a lower-bound estimate, not a certification. The paper supplies no upper bound on  $\lambda_c$ : even  $\lambda_c < 1$  is maintained rather than verified. No tension with the saddle structure arises:  $\Gamma_{\text{endo}}$  is the equilibrium law of motion after the forward-looking variable has been solved out, so the unstable direction of the  $(K, C)$  system is absent from the distributional dynamics.

---

<sup>18</sup>Stepping back one period, two independent histories still agree if both persist ( $\rho_A^2$ ) or both switch ( $(1 - \rho_A)^2$ ); for asymmetric chains,  $q$  is the backward agreement probability of the time-reversed chain.

**Ergodic interpretation** Theorem 2(i) licenses reading Proposition 9 on the ergodic set. Every state observed along a long simulated path is, up to a geometrically small error, the long-history limit of its own realized history. Every pair in the same- $K$  cloud is therefore a pair of long-history limits, inheriting the pathwise bound  $d(\Phi, \Phi') \leq \lambda_c^T \text{diam}(\mathcal{U})$  with  $T$  the realized common-suffix length. This applies at every date, deep within a spell or just after a switch, since the argument runs over histories rather than spells.

The expectation formula, by contrast, is computed under the idealized model of two independent stationary histories, and two caveats attach: within-path pairs are dependent, and conditioning on a common  $K$  is a selection. The closed form is therefore a benchmark for the typical scale of the cloud rather than a bound on its realized diameter.

Theorem 2 leaves  $\lambda_c$  unspecified. The next proposition pins its value near the conditional steady state: the stable eigenvalue of the linearized frozen-regime dynamics, the same object governing the local saddle slope (Proposition 5). This holds under the standard regularity of infinite-dimensional linearization: smoothness, a spectral gap, a stable manifold, and metric-norm agreement. Expressing the rate in terms of aggregate capital requires a fifth condition:  $K$  is  $C^1$  and nondegenerate along the stable direction. Appendix D states the five conditions precisely and explains each briefly.

**Proposition 10** (Local contraction rate on the conditional saddle).

*Near the conditional steady state, the frozen-regime dynamics on the conditional saddle contract distances at the geometric rate  $|\lambda_A^s|$ , where  $\lambda_A^s \in (0, 1)$  is the stable eigenvalue of the linearized frozen-regime dynamics; in terms of aggregate capital,*

$$K_{t+1} - K_A^{cs} = \lambda_A^s (K_t - K_A^{cs}) + o(|K_t - K_A^{cs}|).$$

*Proof.*

See Appendix D. ■

The full statement on the distributional state space, its regularity conditions, and the proof are in Appendix D. Directions orthogonal to the saddle may contract at different rates, so the global rate  $\lambda_c$  in Theorem 2 could exceed  $|\lambda_A^s|$ . The proposition delivers the primitive anchor, the same eigenvalue that drives the slope-ratio diagnostic (Proposition 5) and disciplines the local contraction in Theorem 2 and Proposition 9. A certified uniform upper bound on  $\lambda_c$  is left to future work.

**A sample-based diagnostic for  $\lambda_c$**  The global uniform  $\lambda_c$  in Theorem 2 bounds contraction across a cloud of distributional states at the same  $K$  reached from different histories, which



need not lie on a single conditional saddle. The diagnostic below can show the map contracts no faster than some observed rate; it cannot certify that it contracts at least that fast for every pair of states, which is what the bound needs. Since the across-spell bound uses  $\lambda_c$  as an *upper* bound on the per-step rate, a certified verification of Proposition 9 requires a certified upper bound on  $\lambda_c$ . Such a bound would come via either a Lipschitz constant for  $\Gamma_{\text{endo}}$  on  $\mathcal{U}$  (model-specific) or a statistical confidence bound on sample estimates (standard but tedious in function spaces). This is beyond the scope of the paper and left to future work. Instead I propose a sample-based *diagnostic* that measures the one-step contraction actually exhibited by the computed map; Santos and Peralta-Alva (2005) give conditions under which moments of simulated approximate equilibria converge to their exact counterparts.

The diagnostic draws  $N$  pairs of distributional states at common  $K$  and regime from the simulated equilibrium, applies the frozen-regime map to both members of each pair, and reports  $\hat{\lambda}_c^N := \max_i r_i$  over the one-step ratios  $r_i$ . The ratios are computed in total variation on the joint distribution over assets and employment. The map is non-expansive in that metric by construction, so  $r_i \leq 1$  holds automatically and the diagnostic measures how far below one the observed rates fall. A contraction modulus depends on the metric, not only the topology, so this does not bound the  $\lambda_c$  of Definition 7, and no such inequality is claimed. Substituting it in the across-spell formula illustrates what the formula delivers at an observed rate rather than bounding it.

Under the Krusell and Smith (1998) calibration, quarterly regime persistence is  $\rho_A = \pi_{AA} = 0.875$  (average 8-quarter spells), so  $q = \rho_A^2 + (1 - \rho_A)^2 \approx 0.781$ . On the simulated panel the diagnostic yields  $\hat{\lambda}_c^N \approx 0.999$ , with a median pairwise ratio of 0.982 over some 45,000 same- $K$  pairs. The sample therefore exhibits almost no contraction margin, and at such a rate the benchmark formula of Proposition 9 would leave a factor of 0.997 multiplying  $\text{diam}(\mathcal{U})$ , which is uninformative. The gap from the along-saddle rate is the one Proposition 10 anticipates. The deterministic benchmark saddle contracts at 0.965, but directions off the saddle contract far more slowly, and the uniform rate is set by the slowest of them. The direct quantitative check, comparing the globally solved  $K'$  at each spell boundary against the prediction from tracking only  $(K, A)$ , gives an average discrepancy of  $2.5 \times 10^{-4}$  in log terms (1.3% of one standard deviation of  $\ln K$ ). This measures a mean forecast error in  $K'$  rather than a worst-case distance between distributions, so it is not directly comparable to the uniform distributional bound above; it separately documents that the across-spell error is small in this calibration.



**State reduction, functional form, and block recursivity** The [Krusell and Smith \(1998\)](#) approach is often labeled near-rational ([Akerlof and Yellen, 1985](#)), but this conflates two sources of approximation the conditional-saddle structure separates. Within an orbit, conditioning on  $K$  is exact, and only the functional form of the law of motion approximates. This within-orbit exactness connects the conditional sufficiency of  $K$  to *block recursive equilibrium* (BRE), where aggregate prices and quantities follow from a low-dimensional state ([Menzio and Shi, 2010, 2011](#); [Schaal, 2017](#)). In directed-search environments block recursivity holds by construction; here it emerges spell-wise from the saddle geometry, with prices, quantities, and  $\Phi$  recoverable from  $K$  along a given conditional saddle. Across histories, by contrast, a residual state-reduction error remains, small in this calibration, which the framework identifies rather than assumes away; additional forecasting moments can reduce it, with gain bounded by the across-spell discrepancy. This is where  $K$ -sufficiency is weaker than exact aggregation, in which  $(K, A)$  alone determines the aggregates regardless of history. Recovery from  $K$  also requires knowing which conditional saddle the economy is on, and that saddle is selected by the sequence of regimes that generated the current spell, not by  $(K, A)$  alone. Two different histories can reach the same  $(K, A)$  while carrying different distributions  $\Phi$ , and hence lie on different orbits. These distributions coincide only approximately, and the agreement is closer the more the two histories share their recent regime path (Proposition 9).

**Verifying monotonicity in practice** Theorem 1 and Proposition 8 justify the repeated transition method ([Lee, 2026a](#)), which uses  $K$  as the conditional sufficient statistic in heterogeneous-agent models, with  $K$  indexing the equilibrium state without pinning down the functional form of the law of motion. The framework also yields a practical tool: a cheap check of the underlying monotonicity hypothesis (Assumption 2 with unique  $K$ -stationarity), run on the computed equilibrium itself: a method that assumes scalar-state sufficiency solves first, then certifies the assumption on its own output or flags its failure, rather than leaving it maintained. The check comes in two forms: an *order-preservation check* (Appendix G) delivers within-spell  $K$ -monotonicity from the computed frozen-regime map where that map is scalar, and pathwise per realized spell otherwise. A strictly increasing map with a unique stable rest point converges monotonically, and in the [Krusell and Smith \(1998\)](#) economy next-period aggregate capital is strictly increasing in current capital within each regime (rank correlation 0.9999–1.0000). Independently, primitive bounds on household MPCs and price feedback control side-switching under rational expectations, yielding exact within-spell

$K$ -sufficiency under log utility and Cobb–Douglas production. The global check is verified along a computed continuation rather than from primitives alone, with two conditions maintained explicitly, a pathwise finite-difference bound on aggregate consumption and the residual no-plateau condition (Appendix E). The logic extends to bonds, GHH labor, and multiple predetermined states, its boundary the lumpy-investment economy (Khan and Thomas, 2008; Bachmann et al., 2013) where nullcline invariance fails and monotonicity becomes a testable diagnostic rather than a theorem (Appendix I).

## 4.2 An investment externality: the conditional saddle surface

This second application extends the Section 2.2 benchmark with a second predetermined stock: investment now also builds aggregate knowledge  $H$ , which augments productivity but is not priced by firms. This combines the cumulative-investment learning of Arrow (1962), the nonappropriable knowledge of Romer (1986), and the two-stock propagation of Cooper and Johri (2002).<sup>19</sup> With two predetermined stocks, frozen orbits form a surface. The planner supplies a second surface; under homogeneity both become cones and the saddle reappears in ratios.

**An endogenous-productivity augmentation** Relative to Section 2.2, the only addition is a second stock. Aggregate investment builds a stock of knowledge  $H$  that augments productivity, so effective productivity and the two accumulation equations are

$$\tilde{A}_t := A_t H_t^\eta, \quad K_{t+1} = (1 - \delta)K_t + I_t, \quad H_{t+1} = (1 - \delta_H)H_t + \gamma I_t^\phi, \quad (32)$$

where  $g(I) := \gamma I^\phi$  denotes the knowledge that aggregate investment generates. Because knowledge enters production only through  $\tilde{A}_t$ , output is  $Y_t = \tilde{A}_t K_t^\alpha = A_t K_t^\alpha H_t^\eta$ , the resource constraint  $C_t + I_t = Y_t$  is unchanged, and Section 2's price schedules carry over with  $A_t$  replaced by  $\tilde{A}_t$ . The Section 2 household becomes an atomistic member of a unit continuum: firms take the knowledge stock as given and households its aggregate law of motion, so no private decision internalizes the effect of investment on  $H_{t+1}$ . Only  $A_t$  is exogenous, so freezing the regime still leaves  $H_t$ , and hence  $\tilde{A}_t$ , evolving. The private Euler equation is

---

<sup>19</sup>Benkard (2000) documents organizational forgetting, which motivates the knowledge depreciation  $\delta_H > 0$ . Appendix K gives the supporting evidence, the planner derivation, the surface validation, and the replication code.

therefore the one from Section 2 evaluated at  $\tilde{A}_t$ , and it excludes the knowledge return:

$$\frac{1}{C_t} = \beta \mathbb{E}_t \left[ \frac{1}{C_{t+1}} \{1 + r(K_{t+1}, \tilde{A}_{t+1})\} \right] = \beta \mathbb{E}_t \left[ \frac{1 - \delta + \alpha A_{t+1} K_{t+1}^{\alpha-1} H_{t+1}^\eta}{C_{t+1}} \right]. \quad (33)$$

The stationary calibration satisfies  $\alpha + \eta\phi < 1$ ; the homogeneous limit below sets  $(\eta, \phi) = (1 - \alpha, 1)$ . The quarterly calibration is standard,  $\beta = 0.99$ ,  $\delta = 0.025$ ,  $\alpha = 0.36$ , and  $A \in \{0.99, 1.01\}$  with persistence  $\pi_{AA} = 0.95$ ; knowledge adds  $(\delta_H, \phi, \eta) = (0.10, 0.7, 0.01)$ , with  $\gamma$  normalizing the deterministic competitive  $H^*$  to one (Appendix K). These three are illustrative rather than estimated, the spillover deliberately small, so the geometry rather than the size of the correction carries the argument. The steady-state credit depends on  $(\eta, \phi)$  only through  $m := \eta\phi$ : the planner's unit-investment condition splits into a capital piece and a knowledge piece, the credit is the second, and with  $a = \alpha/[1 - \beta(1 - \delta)]$  and  $b = \delta_H/[\delta[1 - \beta(1 - \delta_H)]]$  it equals  $\bar{\tau} = bm/(a + bm)$ , with elasticity  $1 - \bar{\tau} \approx 0.98$  in the spillover. The magnitudes below therefore scale with the assumed spillover to a close approximation, while the qualitative results do not.

**The planner and the credit** Because investment jointly produces next-period capital and knowledge, the planner carries *two* co-states,  $p = V_K$  and  $q = V_H$ , alongside  $\lambda = 1/C$ :

$$\begin{aligned} \lambda_t &= \beta \mathbb{E}_t[p_{t+1}] + g'(I_t) \beta \mathbb{E}_t[q_{t+1}], & p_t &= \lambda_t \alpha \frac{Y_t}{K_t} + (1 - \delta) \beta \mathbb{E}_t[p_{t+1}], \\ q_t &= \lambda_t \eta \frac{Y_t}{H_t} + (1 - \delta_H) \beta \mathbb{E}_t[q_{t+1}]. \end{aligned} \quad (34)$$

In the standard model, substituting the investment condition  $\lambda_t = \beta \mathbb{E}_t[p_{t+1}]$  into the capital envelope collapses the shadow value of capital to  $p_t = \lambda_t(1 - \delta + \alpha Y_t/K_t)$ . That step fails here: the investment condition carries the extra knowledge term, so the shadow value of installed capital lies below that benchmark by  $(1 - \delta)g'(I_t)\beta \mathbb{E}_t[q_{t+1}]$ , and the knowledge return enters as a *flow* on gross investment, not as a capitalized addition to the price of capital. A Pigouvian investment credit that prices the spillover,

$$\tau_t = \frac{g'(I_t) \beta \mathbb{E}_t[q_{t+1}]}{\lambda_t}, \quad (35)$$

restores efficiency. Because  $I_t$ ,  $q_{t+1}$ , and  $\lambda_t$  all depend on the inherited stocks, the credit is itself a surface over  $(K, H)$  in each regime, the policy counterpart of the two allocation surfaces. At the calibration the deterministic planner steady state lies 3.9 percent above the competitive one in capital and 2.7 percent in knowledge, with  $\bar{\tau} = 0.0242$ . Implemented as a gross-investment credit financed by a lump-sum levy  $\mathcal{T}_t = \tau_t I_t$ , it lowers the effective price of gross individual investment  $i_t := k_{t+1} - (1 - \delta)k_t$  to  $1 - \tau_t$  with a balanced budget each

period. The credited household budget and Euler equation are

$$\begin{aligned} c_t + (1 - \tau_t)i_t &= \frac{\alpha Y_t}{K_t} k_t + w_t - \mathcal{T}_t, \\ (1 - \tau_t)\lambda_t &= \beta \mathbb{E}_t \left[ \lambda_{t+1} \left\{ \frac{\alpha Y_{t+1}}{K_{t+1}} + (1 - \tau_{t+1})(1 - \delta) \right\} \right], \end{aligned} \quad (36)$$

where undepreciated capital is valued at the net-of-credit replacement price. Substituting the planner co-state conditions shows that the state-contingent credit (35) reproduces the planner's allocation exactly; fed back through this credited Euler on the computed planner path, it holds to a residual at solve accuracy (Appendix K).

**The conditional saddle surface** In the computed basin, each pair of initial stocks  $(K_0, H_0)$  launches a one-dimensional frozen-regime orbit  $\mathcal{M}((K_0, H_0), A)$  (agents still price switches). Their closure,  $\mathcal{S}_A = \overline{\bigcup_{(K_0, H_0)} \mathcal{M}((K_0, H_0), A)}$ , is the consumption-policy surface over  $(K, H)$ . Figure 7(a) traces it as orbit fans. The deterministic-equation benchmark has two stable roots against one unstable, so the surface is a genuine saddle.<sup>20</sup> The rest points  $(37.07, 0.983)$  and  $(38.92, 1.017)$  bracket the cloud's capital range,  $K \in [37.10, 38.91]$ . The two stable roots separate the stocks: the fast root, 0.903, is almost purely a knowledge direction at the knowledge depreciation rate  $1 - \delta_H = 0.90$  (half-life about seven quarters); the slow root, 0.965, is capital-dominated (half-life about twenty). Knowledge therefore settles quickly onto the stationary locus  $H = \gamma(\delta K)^\phi / \delta_H$ , which is common to both regimes, and the economy then crawls along it in capital. The ergodic set is correspondingly a thin ribbon about that locus, departing from it by 0.8 percent on average against a 4.3 percent range in  $H$ , and the planner inherits the same structure. Since the mean regime spell is also twenty quarters, the economy never completes the capital adjustment before the regime switches: the rest points are attractors it approaches but does not reach.

**The planner's surface and where it rests** The planner faces the same two stocks, so its frozen dynamics trace a second surface over the same base, itself a saddle with two stable roots against three, counted in the five-variable system  $(K, H, C, q, p)$ : the static consumption condition contributes an infinite root, and eliminating  $C$  leaves two stable against two finite unstable (Appendix K). The two surfaces are not usefully compared point by point, because the economies they describe do not occupy the same states: the

<sup>20</sup>Appendix K reports the counts, eigenvalue moduli, and frozen-map classification: the frozen-map Jacobian is stable at  $B$  and resolution-ambiguous at  $G$ : differencing the interpolated policy surface along a near-neutral direction leaves the slow root's modulus spanning 0.98–1.04. The dedicated frozen  $G$  path converges to the rest point at decay rates 0.94–0.96, establishing attraction without identifying the second local eigenvalue.

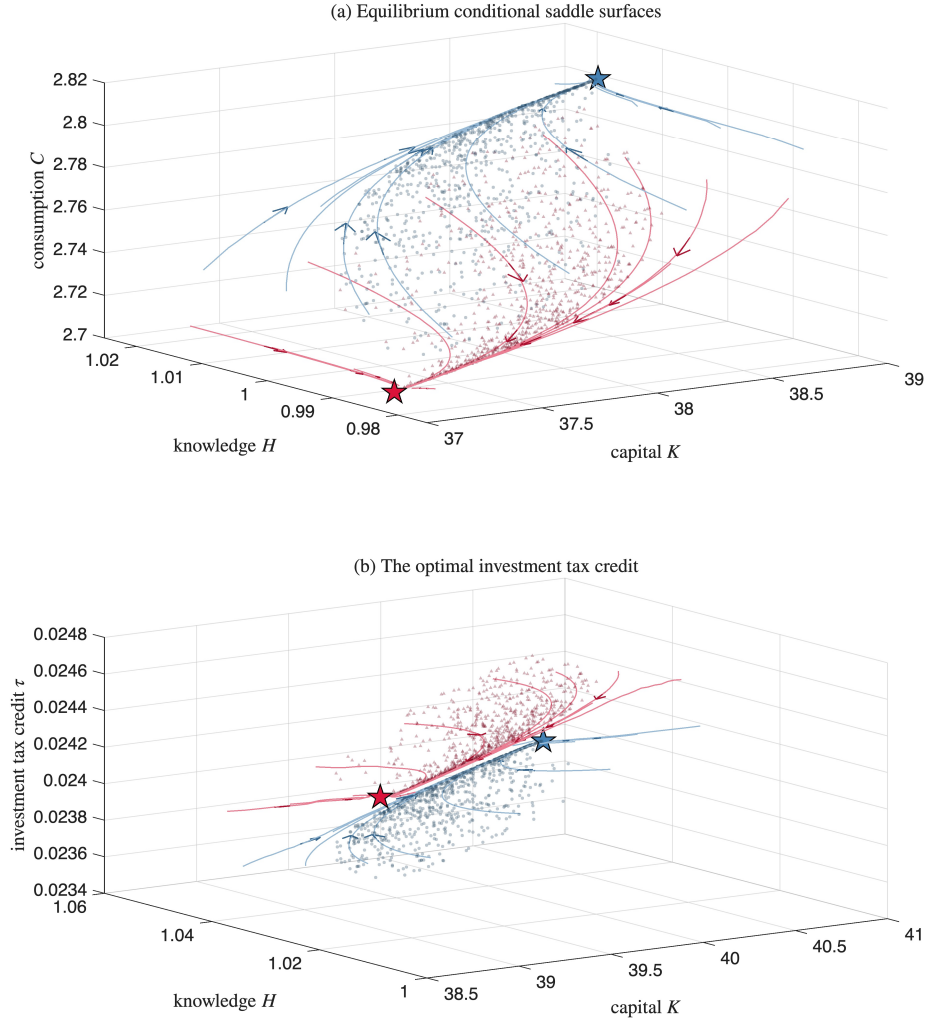


Figure 7: The conditional saddle surface and the optimal investment tax credit.

*Notes:* Panel (a): the good-regime (steel blue) and bad-regime (crimson) equilibrium conditional saddle surfaces, fans of frozen orbits converging into the conditional rest points (stars; Appendix K); arrows mark the direction of motion. Panel (b): the optimal investment tax credit (35) as a surface over the same base, drawn in the same format from frozen planner orbits, with stars at the planner's conditional rest points.

externality moves the whole configuration rather than tilting one surface against a fixed other. What is well defined is where each surface rests. From the same frozen-regime solves, the planner's conditional rest points are  $(38.54, 1.010)$  and  $(40.43, 1.045)$ , against the equilibrium's  $(37.07, 0.983)$  and  $(38.92, 1.017)$ : each attractor is displaced by about 3.9 percent of capital. Since each economy clusters around its own pair, the two recurrent sets

barely meet. The planner's ergodic capital range  $[38.59, 40.43]$  overlaps the equilibrium's  $[37.10, 38.91]$  only over  $[38.59, 38.91]$ , and 12.7 percent of planner dates fall inside the equilibrium's range. Internalizing the spillover therefore raises consumption on average, by 0.6 percent at an essentially equal log standard deviation, though the planner saves more from any state. This compares ergodic distributions and abstracts from the transition, so it is not a welfare gain from adopting the credit.

**The tax credit surface** Figure 7(b) draws the optimal tax credit (35) on the same base as the allocation surfaces. It is state-dependent and countercyclical, averaging 2.45 percent in the bad regime against 2.39 percent in the good, because low investment raises the marginal knowledge return  $g'(I_t)$ . The same force makes it rise with capital, while diminishing returns to knowledge make it fall with  $H$ . A constant credit matches only the deterministic planner steady state and need not implement the planner elsewhere. The two nevertheless nearly coincide here, not because the surface is flat but because the economy travels so little across it. The surface is genuinely tilted, with elasticities of +0.6 in capital and  $-0.5$  in knowledge; the ergodic set spans about four percent in each stock, so  $\tau_t$  stays within  $[0.0237, 0.0247]$ . From a common anchor under identical shocks,  $\bar{\tau} = 0.0242$  therefore tracks the planner within 0.1 percent over eight hundred quarters. Matching the planner's steady state is thus a weaker requirement than implementing its policy along transitions; the credit surface makes the gap explicit.

**Boundary of the construction** The surface exists because the composite returns satisfy  $\alpha + \eta\phi < 1$ , which gives two stable roots for the two predetermined stocks. At equality a unit root appears, and above the boundary only one stable direction remains, so generic two-stock convergence fails. This is a loss of saddle convergence, not sunspot indeterminacy, because this inelastic-labor economy never produces more stable roots than predetermined states. In the exact homogeneous case,  $\phi = 1$  and  $\eta = 1 - \alpha$ , the frozen-regime system is homogeneous of degree one, the surface becomes a cone, and the ratio  $K/H$  is exactly sufficient for the normalized dynamics. This is dimension reduction by symmetry rather than by the monotonicity of Section 4.1, and it holds in all 642 realized spells of that case (Appendix K).

## 5 Concluding remarks

Conditional saddle paths, equilibrium trajectories under frozen exogenous regimes, extend the geometric analyses of deterministic phase diagrams to stochastic environments. In particular, this paper highlights three distinguished properties of the conditional saddle paths. Slopes capture state dependence: differential tilt reflects asymmetric shock responses, and distance along the saddle shapes the response even at nearly equal slopes. Monotonicity delivers dimension reduction: a strictly monotone aggregate exactly indexes the state within each frozen-regime spell, the structural reason scalar-state methods work. With a second predetermined stock, the path becomes a surface, and its gap from the planner's surface visualizes the distortion and the policy that closes it: an investment credit, itself a surface over the two stocks, state-dependent and countercyclical.

More broadly, questions that have required separate numerical results, such as when a scalar state suffices, how responses depend on a position in the state space, and what the role of anticipation and the welfare implications are, become legible in a single geometric representation. This framework opens a new pathway for stochastic equilibrium analysis: what the geometry makes visible, the data can measure and policy can act on. Its slope and timing predictions are moments that state-dependent local projections can estimate from the data, disciplining the primitives behind them, and its policy surfaces extend to stabilization rules that condition on the aggregate state.

## References

- ABEL, A. B. AND J. C. EBERLY (1996): "Optimal Investment with Costly Reversibility," *The Review of Economic Studies*, 63, 581–593.
- ACHDOU, Y., J. HAN, J.-M. LASRY, P.-L. LIONS, AND B. MOLL (2022): "Income and Wealth Distribution in Macroeconomics: A Continuous-Time Approach," *The Review of Economic Studies*, 89, 45–86.
- AIYAGARI, S. R. (1994): "Uninsured Idiosyncratic Risk and Aggregate Saving," *The Quarterly Journal of Economics*, 109, 659–684.
- AKERLOF, G. A. AND J. L. YELLEN (1985): "Can Small Deviations from Rationality Make Significant Differences to Economic Equilibria?" *American Economic Review*, 75, 708–720.
- ANDREASEN, M. M., J. FERNÁNDEZ-VILLVERDE, AND J. F. RUBIO-RAMÍREZ



- (2018): “The Pruned State-Space System for Non-Linear DSGE Models: Theory and Empirical Applications,” *The Review of Economic Studies*, 85, 1–49.
- ARNOLD, L. (1998): *Random Dynamical Systems*, Springer Monographs in Mathematics, Berlin, Heidelberg: Springer Berlin Heidelberg.
- ARROW, K. J. (1962): “The Economic Implications of Learning by Doing,” *Review of Economic Studies*, 29, 155–173.
- AUCLERT, A., B. BARDÓCZY, M. ROGNLIE, AND L. STRAUB (2021): “Using the Sequence-Space Jacobian to Solve and Estimate Heterogeneous-Agent Models,” *Econometrica*, 89, 2375–2408.
- AUERBACH, A. J. AND Y. GORODNICHENKO (2012): “Measuring the Output Responses to Fiscal Policy,” *American Economic Journal: Economic Policy*, 4, 1–27.
- AZINOVIC, M., L. GAEGAUF, AND S. SCHEIDEGGER (2022): “Deep Equilibrium Nets,” *International Economic Review*, 63, 1471–1525.
- AZINOVIC-YANG, M. AND J. ŽEMLIČKA (2025): “Deep Learning in the Sequence Space,” Working paper, arXiv:2509.13623.
- BACHMANN, R., R. J. CABALLERO, AND E. M. R. A. ENGEL (2013): “Aggregate Implications of Lumpy Investment: New Evidence and a DSGE Model,” *American Economic Journal: Macroeconomics*, 5, 29–67.
- BALEY, I. AND A. BLANCO (2019): “Firm Uncertainty Cycles and the Propagation of Nominal Shocks,” *American Economic Journal: Macroeconomics*, 11, 276–337.
- BARRO, R. J. AND X. SALA-I-MARTIN (1992): “Convergence,” *Journal of Political Economy*, 100, 223–251.
- BASU, S. AND B. BUNDICK (2017): “Uncertainty Shocks in a Model of Effective Demand,” *Econometrica*, 85, 937–958.
- BENKARD, C. L. (2000): “Learning and Forgetting: The Dynamics of Aircraft Production,” *American Economic Review*, 90, 1034–1054.
- BERGER, D., K. MILBRADT, F. TOURRE, AND J. VAVRA (2021): “Mortgage Prepayment and Path-Dependent Effects of Monetary Policy,” *American Economic Review*, 111, 2829–78.
- BERGER, D. AND J. VAVRA (2015): “Consumption Dynamics During Recessions,” *Econometrica*, 83, 101–154.
- BEWLEY, T. (1986): “Stationary Monetary Equilibrium with a Continuum of Independently Fluctuating Consumers,” in *Contributions to Mathematical Economics in Honor of Gérard*

- Debreu*, ed. by W. Hildenbrand and A. Mas-Colell, North-Holland, 79–102.
- BHANDARI, A., T. BOURANY, D. EVANS, AND M. GOLOSOV (2023): “A Perturbational Approach for Approximating Heterogeneous Agent Models,” NBER Working Paper 31744.
- BIANCHI, F. (2016): “Methods for Measuring Expectations and Uncertainty in Markov-Switching Models,” *Journal of Econometrics*, 190, 79–99.
- BILAL, A. (2023): “Solving Heterogeneous Agent Models with the Master Equation,” Working Paper 31103, National Bureau of Economic Research.
- BLANCHARD, O. J. AND C. M. KAHN (1980): “The Solution of Linear Difference Models under Rational Expectations,” *Econometrica*, 48, 1305–1311.
- BLOOM, N., M. FLOETOTTO, N. JAIMOVICH, I. SAPORTA-EKSTEN, AND S. J. TERRY (2018): “Really Uncertain Business Cycles,” *Econometrica*, 86, 1031–1065.
- BOPPART, T., P. KRUSELL, AND K. MITMAN (2018): “Exploiting MIT shocks in heterogeneous-agent economies: the impulse response as a numerical derivative,” *Journal of Economic Dynamics and Control*, 89, 68–92.
- BOURANY, T. (2018): “Wealth Distribution over the Business Cycle: A Mean-Field Game with Common Noise,” Master’s thesis, UPMC-Sorbonne Université, supervised by Yves Achdou, Paris-Diderot University (LJLL).
- BROCK, W. A. AND L. J. MIRMAN (1972): “Optimal Economic Growth and Uncertainty: The Discounted Case,” *Journal of Economic Theory*, 4, 479–513.
- BUERA, F. J., J. P. KABOSKI, M. MESTIERI, AND D. G. O’CONNOR (2024): “The Stable Transformation Path,” Working paper.
- CAO, D. (2020): “Recursive equilibrium in Krusell and Smith (1998),” *Journal of Economic Theory*, 186, 104978.
- CARROLL, C. D. (1997): “Buffer-Stock Saving and the Life Cycle/Permanent Income Hypothesis,” *The Quarterly Journal of Economics*, 112, 1–55.
- CASS, D. (1965): “Optimum Growth in an Aggregative Model of Capital Accumulation,” *The Review of Economic Studies*, 32, 233–240.
- CHILDERS, R. L. (2018): “Solution of Rational Expectations Models with Function Valued States,” Working paper.
- CHUNG, H., T. DAVIG, AND E. M. LEEPER (2007): “Monetary and Fiscal Policy Switching,” *Journal of Money, Credit and Banking*, 39, 809–842.
- COOPER, R. AND A. JOHRI (2002): “Learning-by-Doing and Aggregate Fluctuations,”

- Journal of Monetary Economics*, 49, 1539–1566.
- COOPER, R. W. AND J. C. HALTIWANGER (2006): “On the Nature of Capital Adjustment Costs,” *The Review of Economic Studies*, 73, 611–633.
- DEATON, A. (1991): “Saving and Liquidity Constraints,” *Econometrica*, 59, 1221–1248.
- DECHERT, W. D. AND K. NISHIMURA (1983): “A Complete Characterization of Optimal Growth Paths in an Aggregated Model with a Non-Concave Production Function,” *Journal of Economic Theory*, 31, 332–354.
- DEN HAAN, W. J. (2010): “Assessing the accuracy of the aggregate law of motion in models with heterogeneous agents,” *Journal of Economic Dynamics and Control*, 34, 79–99.
- DEN HAAN, W. J. AND A. MARCET (1990): “Solving the Stochastic Growth Model by Parameterizing Expectations,” *Journal of Business & Economic Statistics*, 8, 31–34.
- DEN HAAN, W. J. AND P. RENDAHL (2010): “Solving the incomplete markets model with aggregate uncertainty using explicit aggregation,” *Journal of Economic Dynamics and Control*, 34, 69–78.
- FERNÁNDEZ-VILLAVÉRDE, J., S. HURTADO, AND G. NUÑO (2023): “Financial Frictions and the Wealth Distribution,” *Econometrica*, 91, 869–901.
- FOERSTER, A., J. F. RUBIO-RAMÍREZ, D. F. WAGGONER, AND T. ZHA (2016): “Perturbation Methods for Markov-Switching Dynamic Stochastic General Equilibrium Models,” *Quantitative Economics*, 7, 637–669.
- GORMAN, W. M. (1961): “On a Class of Preference Fields,” *Metroeconomica*, 13, 53–56.
- HAN, J., Y. YANG, AND W. E (2026): “DeepHAM: A Global Solution Method for Heterogeneous Agent Models with Aggregate Shocks,” *Quantitative Economics*, 17, 297–341.
- HUGGETT, M. (1993): “The Risk-Free Rate in Heterogeneous-Agent Incomplete-Insurance Economies,” *Journal of Economic Dynamics and Control*, 17, 953–969.
- JUDD, K. L., L. MALIAR, AND S. MALIAR (2011): “Numerically stable and accurate stochastic simulation approaches for solving dynamic economic models,” *Quantitative Economics*, 2, 173–210.
- KAMIHIGASHI, T. (2003): “Necessity of transversality conditions for stochastic problems,” *Journal of Economic Theory*, 109, 140–149.
- (2005): “Necessity of the transversality condition for stochastic models with bounded or CRRA utility,” *Journal of Economic Dynamics and Control*, 29, 1313–1329.

- KAPLAN, G., B. MOLL, AND G. L. VIOLANTE (2018): “Monetary Policy According to HANK,” *American Economic Review*, 108, 697–743.
- KAPLAN, G. AND G. L. VIOLANTE (2014): “A Model of the Consumption Response to Fiscal Stimulus Payments,” *Econometrica*, 82, 1199–1239.
- KHAN, A. AND J. K. THOMAS (2008): “Idiosyncratic Shocks and the Role of Nonconvexities in Plant and Aggregate Investment Dynamics,” *Econometrica*, 76, 395–436.
- KOOP, G., M. PESARAN, AND S. M. POTTER (1996): “Impulse response analysis in nonlinear multivariate models,” *Journal of Econometrics*, 74, 119–147.
- KOOPMANS, T. (1963): “On the Concept of Optimal Economic Growth,” Cowles Foundation Discussion Papers 163, Cowles Foundation for Research in Economics, Yale University.
- KRUSELL, P. AND A. A. SMITH, JR. (1998): “Income and Wealth Heterogeneity in the Macroeconomy,” *Journal of Political Economy*, 106, 867–896.
- KUBLER, F. AND K. SCHMEDDERS (2003): “Stationary Equilibria in Asset-Pricing Models with Incomplete Markets and Collateral,” *Econometrica*, 71, 1767–1793.
- LEE, H. (2026a): “A Global Dynamic Nonlinear Solution Framework and the Repeated Transition Method,” Working paper.
- (2026b): “Striking While the Iron Is Cold: Fragility after a Surge of Lumpy Investments,” *Journal of the European Economic Association*, forthcoming.
- LUCAS, ROBERT E., J. (1987): *Models of Business Cycles*, Oxford: Basil Blackwell.
- MALIAR, L., S. MALIAR, AND P. WINANT (2021): “Deep learning for solving dynamic economic models,” *Journal of Monetary Economics*, 122, 76–101.
- MARCET, A. (1988): “Solving nonlinear stochastic models by parameterizing expectations,” Manuscript. Pittsburgh: Carnegie Mellon Univ.
- MELCANGI, D. (2024): “Firms’ precautionary savings and employment during a credit crisis,” *American Economic Journal: Macroeconomics*, 16, 356–386.
- MENZIO, G. AND S. SHI (2010): “Block Recursive Equilibria for Stochastic Models of Search on the Job,” *Journal of Economic Theory*, 145, 1453–1494.
- (2011): “Efficient Search on the Job and the Business Cycle,” *Journal of Political Economy*, 119, 468–510.
- MIAO, J. (2006): “Competitive equilibria of economies with a continuum of consumers and aggregate shocks,” *Journal of Economic Theory*, 128, 274–298.
- NEFTCI, S. N. (1984): “Are Economic Time Series Asymmetric over the Business Cycle?”

- Journal of Political Economy*, 92, 307–328.
- PETROSKY-NADEAU, N., L. ZHANG, AND L.-A. KUEHN (2018): “Endogenous Disasters,” *American Economic Review*, 108, 2212–45.
- PIZZINELLI, C., K. THEODORIDIS, AND F. ZANETTI (2020): “State Dependence in Labor Market Fluctuations,” *International Economic Review*, 61, 1027–1072.
- PROEHL, E. (2025a): “Approximating Equilibria with Ex-Post Heterogeneity and Aggregate Risk,” Working paper.
- (2025b): “Existence and Uniqueness of Recursive Equilibria with Aggregate and Idiosyncratic Risk,” Working paper.
- RAMSEY, F. P. (1928): “A Mathematical Theory of Saving,” *The Economic Journal*, 38, 543–559.
- REDDY, P. V., J. M. SCHUMACHER, AND J. C. ENGWERDA (2020): “Analysis of Optimal Control Problems for Hybrid Systems with One State Variable,” *SIAM Journal on Control and Optimization*, 58, 3262–3292.
- REITER, M. (2010): “Solving the incomplete markets model with aggregate uncertainty by backward induction,” *Journal of Economic Dynamics and Control*, 34, 28–35.
- ROGNLIE, M., A. SHLEIFER, AND A. SIMSEK (2018): “Investment Hangover and the Great Recession,” *American Economic Journal: Macroeconomics*, 10, 113–153.
- ROMER, P. M. (1986): “Increasing Returns and Long-Run Growth,” *Journal of Political Economy*, 94, 1002–1037.
- SANTOS, M. S. AND A. PERALTA-ALVA (2005): “Accuracy of Simulations for Stochastic Dynamic Models,” *Econometrica*, 73, 1939–1976.
- SCHAAL, E. (2017): “Uncertainty and Unemployment,” *Econometrica*, 85, 1675–1721.
- SCHENK-HOPPÉ, K. R. (2001): “Random Dynamical Systems in Economics,” *Stochastics and Dynamics*, 1, 63–83.
- SCHENK-HOPPÉ, K. R. AND B. SCHMALFUB (2001): “Random fixed points in a stochastic Solow growth model,” *Journal of Mathematical Economics*, 36, 19–30.
- SKIBA, A. K. (1978): “Optimal Growth with a Convex–Concave Production Function,” *Econometrica*, 46, 527–539.
- SOLOW, R. M. (1956): “A Contribution to the Theory of Economic Growth,” *The Quarterly Journal of Economics*, 70, 65–94.
- TALLARINI, THOMAS D., J. (2000): “Risk-Sensitive Real Business Cycles,” *Journal of Monetary Economics*, 45, 507–532.

- TENREYRO, S. AND G. THWAITES (2016): “Pushing on a String: US Monetary Policy Is Less Powerful in Recessions,” *American Economic Journal: Macroeconomics*, 8, 43–74.
- VAVRA, J. (2014): “Inflation Dynamics and Time-Varying Volatility: New Evidence and an Ss Interpretation,” *The Quarterly Journal of Economics*, 129, 215–258.
- WAGENER, F. (2003): “Skiba Points and Heteroclinic Bifurcations, with Applications to the Shallow Lake System,” *Journal of Economic Dynamics and Control*, 27, 1533–1561.
- WALSH, K. J. AND E. R. YOUNG (2024): “Equilibrium Multiplicity in Aiyagari and Krusell-Smith,” Federal Reserve Bank of Cleveland Working Paper 24-13.
- WINBERRY, T. (2018): “A method for solving and estimating heterogeneous agent macro models,” *Quantitative Economics*, 9, 1123–1151.
- (2021): “Lumpy Investment, Business Cycles, and Stimulus Policy,” *American Economic Review*, 111, 364–396.
- YANNAKOPOULOS, A. N. (2011): “Stochastic Saddle Paths and Economic Theory,” in *Dynamics, Games and Science II*, ed. by M. M. Peixoto, A. A. Pinto, and D. A. Rand, Berlin, Heidelberg: Springer Berlin Heidelberg, 735–752.
- ZHANG, L. (2005): “The Value Premium,” *The Journal of Finance*, 60, 67–103.